University of Central Florida
STARS

Electronic Theses and Dissertations, 2020-2023

2022

Law Enforcement Motivations for Establishing Real-Time Crime Centers: A Mixed Model Study

Frank Linton
University of Central Florida

Part of the Criminology and Criminal Justice Commons
Find similar works at: https://stars.library.ucf.edu/etd2020
University of Central Florida Libraries http://library.ucf.edu

This Masters Thesis (Open Access) is brought to you for free and open access by STARS. It has been accepted for inclusion in Electronic Theses and Dissertations, 2020-2023 by an authorized administrator of STARS. For more information, please contact STARS@ucf.edu.

STARS Citation
Linton, Frank, "Law Enforcement Motivations for Establishing Real-Time Crime Centers: A Mixed Model Study" (2022). *Electronic Theses and Dissertations, 2020-2023*. 1244.
https://stars.library.ucf.edu/etd2020/1244

LAW ENFORCEMENT MOTIVATIONS FOR ESTABLISHING REAL-TIME CRIME CENTERS: A MIXED MODEL STUDY

by

FRANK S. LINTON
B.S. Florida State University, 2002
B.A. Florida State University, 2002

A thesis submitted in partial fulfillment of the requirements
for the degree of Master of Science
in the Department of Criminal Justice
in the College of Community Innovation and Education
at the University of Central Florida
Orlando, Florida

Summer Term
2022

Major Professors: Roberto Potter & Gail Humiston

© 2022 Frank S. Linton

ABSTRACT

Law enforcement agencies across the United States are establishing real-time crime centers (RTCCs) at a rapid rate, yet research exploring the decisions to establish them is lacking. This master's thesis therefore attempts to fill that void by using a mixed methods approach to examine whether adaptive organizational theories, consisting of contingency theory (Lawrence & Lorsch, 1967; Woodward, 1965), resource dependency theory (Pfeffer & Salancik, 1978), and institutional theory (DiMaggio & Powell 1991; Meyer & Rowan, 1977), can be used to explain a law enforcement agency's decision to set up an RTCC. This cross-sectional mixed methods design comprised two different studies (Study 1 & Study 2). Study 1 consisted of a content analysis of newspaper articles (N = 235) that listed the terms "real-time crime center" or "real time crime center". The content analysis revealed that both resource dependency theory and contingency theory were likely factors for a law enforcement agency's decision to establish an RTCC. Study 2 included interviews of agency leaders of Florida law enforcement departments that had established an RTCC (N = 11) and those that had not (N = 2). A qualitative analysis of the interviews with agency leaders indicates that traditional organizational theories may not fully explain law enforcement agencies' adoption of RTCCs. The interviews partially confirmed the content analysis of newspaper articles while revealing a theme of front-line and mid-management influence on decision-making. This study found that internal lower-ranking "champions," not external forces, are the primary reason law enforcement agencies are establishing RTCCs, a term known as institutional entrepreneurship (DiMaggio 1988; Eisenstadt, 1980). This study is important because it provides a foundation for explaining why law

enforcement agencies choose to set up RTCCs, a law enforcement organizational element whose establishment across the United States does not appear to be slowing anytime soon. This study also highlights an aspect of institutional theory not found in the criminal justice literature. Lastly, limitations and future directions for research are discussed.

This thesis is dedicated to our beloved and faithful companion Trigger. When I began this journey, I had no idea the number of times schoolwork would cause me to work late into the night or into the very early mornings. As was often the case, I would experience self-doubt that such a task was possible, or frustration would creep in at the progress of an assignment. Your constant presence during those times always seemed to provide the boost needed to overcome the doubt and provided the encouragement to just keep going. Although we had to say good-bye at the start of this project, thank you for keeping me company during all those sleepless nights and know that you will forever be in our hearts.

ACKNOWLEDGMENTS

This endeavor would not have been possible without my committee members, particularly Drs. Roberto Potter and Gail Humiston. The patience, knowledge, understanding, and support you have offered over the past three semesters are nothing short of spectacular. Let me confess that I started this journey without having an in-depth understanding of what was really in store for me. I am sincerely grateful for every exchange we had via Zoom call, email, and text message and during visits along this journey.

I would also like to thank all the representatives from the various law enforcement agencies who took time away from their busy schedules to have a conversation with me about their real-time crime centers (RTCCs). The paths that you are forging and the changes you are making in the law enforcement community will hopefully be recognized one day.

Finally, I would be remiss if I did not thank my family. My parents have consistently encouraged me to better myself through education and to constantly strive to challenge myself. For that, you deserve a lot of the credit, especially for instilling in me the drive and focus to attempt such a feat. Finally, I thank my wife Christina. I owe you much of the credit for this accomplishment. For the past two years you did so much that allowed me to truly focus such a daunting task as this, and I can honestly say that accomplishing this feat would not have been possible without you. I am so fortunate and grateful that you agreed to join me on this journey, and we get to experience this accomplishment together. The best part is, now that I am done, you can have your dining room table back.

TABLE OF CONTENTS

LIST OF TABLES ix

LIST OF ACRONYMS OR ABBREVIATIONS x

CHAPTER 1: INTRODUCTION 1

CHAPTER 2: LITERATURE REVIEW 5

Approaches to Proactive Policing 5

Intelligence Led Policing 5

COMPSTAT 6

Contingency Theory 10

Resource Dependency Theory 12

Interorganizational Relationships 14

Asymmetry 15

Reciprocity 15

Efficiency 16

Stability 16

Legitimacy 17

Institutional Theory 17

Summary 21

CHAPTER 3: METHODS AND DATA 22

Research Design 22

Study 1 23

Study 2 29

Findings 34

Summary 48

CHAPTER 4: DISCUSSION 49

Limitations 60

CHAPTER 5: SUMMARY AND CONCLUSION 62

Future Research 63

APPENDIX A: LIST OF TABLES 64

APPENDIX B: SURVEY INSTRUMENT 67

APPENDIX C: IRB APPROVAL LETTER 73

REFERENCES 76

LIST OF TABLES

Table 1 Summary of Adaptive Organizational Theories 65

Table 2 Study 2 Description of Survey Recipients and Respondents..... 66

LIST OF ACRONYMS OR ABBREVIATIONS

Abbreviation	Meaning	Page
COMPSTAT	Computerized statistics or computer comparison statistics	7
ILP	Intelligence-led Policing	4
IOR	Interorganizational Relationship	11
IRB	Institutional Review Board	32
LPR	License Plate Reader	27
MOU	Memorandum of Understanding	54
NIBIN	National Integrated Ballistic Information Network	37
NYPD	New York Police Department	1
RTCC	Real-Time Crime Center	1
SPSS	Statistical Package for the Social Sciences	26

USAI Urban Areas Security Initiative 45

CHAPTER 1: INTRODUCTION

In 2005, the New York Police Department (NYPD) created a new section within the police department called a real-time crime center (RTCC). In this new center, NYPD would leverage the technological infrastructure implemented after the September 11 terrorist attacks and repurpose the technology to detect crimes in progress. Since the founding of the original NYPD RTCC, the pace at which law enforcement agencies have been establishing RTCCs seems to be increasing, yet there has been almost no research to determine the organizational factors behind why agencies are choosing to establish RTCCs.

Researchers have conceptualized organizational change along two broad categories: process and content. Process describes how changes occur. Content describes how the organization changed. From these concepts, two organizational schools of thought have developed. The first is based on the idea that organizations change through a process of natural selection. When environmental change occurs, some organizations will fail and quickly be replaced by newer organizations that are more in tune with the current environment. Eventually, these organizations will also fail and be replaced by more modern organizations. This process continually repeats itself. Popular theories associated with this explanation of organizational change include organizational ecology (Hannan & Freeman, 1977) and evolutionary economics (Nelson & Winter, 2004). The other popular school of thought is that instead of failing, organizations will adapt to their new environment, a process referred to as organizational adaption.

Organizational adaption is built on the idea that managers can assess their environment and look for opportunities that will be successful, thus increasing the chances of their

organization's survival (Sarta et al., 2021). Organizational adaption looks at purposeful change between the organization and the environment and does not look at change from a strategic perspective (Sarta et al., 2021). Popular theories falling under the adaptation camp include transaction cost economics (Williamson, 1975), resource dependency theory (Pfeffer & Salancik, 2003), institutional theory (DiMaggio & Powell, 1991; Meyer & Rowan, 1977), and contingency theory (Lawrence & Lorsch, 1967; Woodward et al., 1965). The theories associated with organizational adaptation are those that will be examined as part of this study to understand why Florida law enforcement agencies are establishing RTCCs.

Understanding why law enforcement agencies are changing and establishing RTCCs is important because RTCCs appear to incorporate a new shift in policing. This shift seems to use technology that focuses on identifying crimes as they occur and data mining intelligence sources to provide as much information as possible as quickly as possible to responding officers. Therefore, this study will attempt to answer the following questions:

1) Is responding to and investigating crime in "real time" the primary reason why Florida law enforcement agencies are creating RTCCs?

2) Did external factors (e.g., elected officials, clergy, judges, community leaders, etc.) play a role in the Florida law enforcement agency's decision to establish an RTCC?

3) Did internal factors play a role in the Florida law enforcement agency's decision to establish an RTCC?

4) Do RTCCs stand alone, or if they interact with organizations, with whom do they interact and how frequently?

It is important to identify the internal and external factors, if any, causing Florida law enforcement agencies to establish RTCCs because RTCCs may represent an innovative form of policing. This new form of policing may be a departure from the two traditional forms, namely reactive and proactive policing.

How a law enforcement agency establishes its RTCC could determine whether it is breaking from the two traditional forms of proactive and reactive policing or using the RTCC to enhance them. In reactive policing, the three main functions of law enforcement include routine patrol, immediate response to calls, and follow-up investigations (Cordner & Sheehan, 1999).

On the opposite end of the spectrum from reactive policing is proactive policing. Proactive policing can be defined as "all policing strategies that have as one of their goals the prevention or reduction of crime and disorder and that are not reactive in terms of focusing primarily on uncovering ongoing crime or on investigating or responding to crimes once they have occurred" (National Academy of Sciences, 2018, pp. 1, 30). Developed in the 1960s in response to rising crime rates, social unrest, and questions regarding whether current police practices were effective, proactive policing consists of four approaches (Braga & Weisburd, 2006; Bundy, 1970; Kelling & Coles, 1996; Weisburd et al., 2021). The four approaches include place-based approaches, problem-solving approaches, community-based approaches, and person-focused approaches (Weisburd et al., 2019). Of these proactive policing approaches, place-based approaches appear to be most closely aligned with the concepts associated with RTCCs. In place-based approaches, law enforcement seeks to focus resources on certain locations within the department's jurisdiction. The emphasis on a particular location allows the police department to concentrate on the causes of crime and deploy limited resources as effectively as possible (Braga & Weisburd, 2006; Sherman & Weisburd, 1995; Weisburd et al., 2008). Popular police strategies

associated with place-based approaches include using closed-circuit cameras, hot spots policing, and predictive policing (Weisburd et al., 2008).

When an agency engages in proactive policing, a significant shift occurs because in reactive policing, citizens determine when to involve the police, while in proactive policing, law enforcement determines the placement of resources, patrol areas, and other enforcement activities without the input from citizens (Waddington, 1993). Popular examples of proactive policing include intelligence-led policing (ILP) and COMPSTAT.

In responding to crimes in progress, initiated by RTCCs, the traditional models of reactive or proactive policing may not apply, so understanding the organizational decisions behind the creation of RTCCs is important as this may be the future of policing in the United States. When RTCCs are utilized, officers are still notified of crimes, but RTCCs can begin data mining for intelligence, often before officers are dispatched. This assistance could consist of photos or videos of suspects, photos of stolen vehicles, or other analytical assistance designed to aid the patrol officers with the call they are currently investigating.

This study aims to explain the history of intelligence-led policing (ILP) and COMPSTAT and how both these proactive policing models laid the foundation for the creation of RTCCs. Popular organizational theories will be compared and used to explain possible reasons as to why law enforcement agencies are creating RTCCs. The paper will discuss how the data were collected and how the challenges of not having a substantial body of prior research on RTCCs were overcome to formulate interview questions. Results of the analysis of the data will be presented and discussed. Finally, recommendations for future research will be offered with the goal of expanding the body of knowledge associated with RTCCs.

CHAPTER 2: LITERATURE REVIEW

Approaches to Proactive Policing

Intelligence Led Policing

The history of ILP can be traced to the early 1980s and 1990s United Kingdom, where police agencies were forced to find creative ways to tackle increased crime threats, including domestic terrorism threats and pressure to decrease crimes while being fiscally conservative (Ratcliffe, 2016). After the September 11 attacks, ILP was considered the most recent trend to diffuse throughout federal, state and local police departments across the United States (Oliver, 2006; Ratcliffe, 2016).

The central tenets of ILP were adopted and refined by the Kent Police Department into what we know today as ILP (Ratcliffe, 2016). The Kent model consists of several tenets including focusing on the most serious criminal offenders. With ILP, management decisions become much clearer regarding which offenders require increased attention (Audit Commission, 1993). ILP also allows law enforcement supervisors to rapidly identify crimes that warrant further investigation. Deploying staff to screen calls and reduce duplication enables investigators to focus on crimes that truly warrant increased attention and screen out those offering no apparent benefit to the overall mission (Heaton, 2000). Since ILP puts such a heavy emphasis on supervisors to promptly identify crimes and criminals that warrant additional attention, supervisors benefit from having a complete picture of the crimes and criminals as soon as possible. As a result, ILP puts a greater emphasis on intelligence-gathering investigative techniques, such as surveillance and the use of informants. Surveillance and intelligence received from informants will often fill in the gaps of the picture, thus allowing supervisors to make the

most informed decision possible regarding which crimes require further attention. Finally, when a law enforcement agency subscribes to the ILP model, the intelligence function of the agency must be at the core of all decision-making (Ratcliffe, 2016).

At its core, ILP is about reducing crime associated with organized groups or prolific offenders, but it does not change the way this task is performed. However, ILP does change how law enforcement agencies use data to become smarter in the allocation of resources and organizational priorities (Shearing & Wood, 2007). While ILP is offender-specific, COMPSTAT, another proactive model of policing, focuses on crime events and police accountability (Ratcliffe, 2016).

COMPSTAT

COMPSTAT was created in 1994 by NYPD Commissioner William Bratton, and it was hailed as a technological advancement in crime fighting that combined crime analysis and geographic information systems with cutting-edge management principles (Willis et al., 2007). While there is no universal meaning of the term COMPSTAT, most people believe it means *computerized statistics* or *computer comparison statistics* (Maple & Mitchell, 1999; Vito, et al., 2005; Walsh, 2005). The main purpose of COMPSTAT was to bring accountability to NYPD's 76 police commanders (Magers, 2004), and when New York's major crimes fell by 50% from 1993 to 1998, COMPSTAT was heralded as the reason (Walsh, 2001).

COMPSTAT consisted of four principles that empowered law enforcement commanders to make changes that were responsive to emerging crime trends. The four principles were (1) accurate and timely information available to all department members; (2) focused law enforcement tactics for a specific crime problem; (3) rapid, focused deployment of law

enforcement officers to address the specific crime problem; and (4) continual assessment to ascertain whether the applied strategies produced the intended effect (McDonald, 2004; Safir, n.d., as citied in Willis et al., 2007). Unlike ILP that did not change the way law enforcement agencies performed their tasks, COMPSTAT compelled police commanders to understand ongoing crime problems in a way that allowed them to implement tactics to effectively combat them. COMPSTAT shifted the focus from crime offenders to crime events (Ratcliffe, 2008). This aspect of COMPSTAT meant failure in successfully dealing with crime problems, leading to a situation where the commanders who succeeded kept their jobs and those who did not adequately address the crime problems were replaced (Vito & Walsh, 2005). COMPSTAT was unique in that crime problems were addressed by mid-level managers instead of patrol officers as in other proactive policing models (Ratcliffe, 2008). Ratcliffe (2008) argues that COMPSTAT is easy to define and adopt since most of the changes occur at the managerial level, thus not requiring a large cultural change at the patrol level.

COMPSTAT was hailed as a transformative shift in policing that was designed to overcome the bureaucratic dysfunction often seen in police departments. The excitement that surrounded COMPSTAT quickly came to the attention of police leaders and policymakers across the country, ensuing a rapid diffusion across law enforcement agencies (Henry, 2003). Despite the positive attention COMPSTAT received, at least one study indicates that agencies did not adopt COMPSTAT to improve their effectiveness as law enforcement agencies but rather to appear progressive and successful (Willis et al., 2007). While ILP and COMPSTAT provide law enforcement managers and supervisors data to explain the types of crimes that are occurring and who is committing them, they do little in assisting patrol officers with investigating crimes in

“real time” or improving officer safety. RTCCs, however, do seem to augment at least some aspects of COMPSTAT and ILP.

RTCCs seem to rely heavily on the use of CCTV cameras, which is analogous to the using surveillance to gather intelligence on criminals and criminal organizations. However, an RTCC does not appear to be a mechanism for informing supervisors which criminals or criminal organizations the agency should direct its resources toward. Additionally, an RTCC does not seem to be a mechanism designed to collect and analyze puzzle pieces that could give law enforcement command staff reasons why crime is occurring or the best way to deal with a particular crime problem.

RTCCs appear to be designed to deal with crimes in progress and once they are over, staff move on to the next call. RTCCs exhibit both similarities and differences with the COMPSTAT model. A central feature of RTCCs is getting accurate and timely information to officers responding to calls. This information is gained from CCTV cameras and other intelligence from law enforcement databases regarding crimes in progress. In this sense, RTCC and COMPSTAT share the feature of providing timely and accurate information to first-line officers. However, an RTCC does not seem to be an accountability mechanism or a proactive model attempting to move law enforcement assets around based on a specific crime problem. Instead, once an RTCC detects a crime or is notified of a crime, RTCC employees can search massive databases that hold large volumes of information and look for information that could assist with the crimes being investigated. The data searches are paired with sensor data, including closed-circuit camera footage, license plate scanners, and gunshot detection warning systems. All this information is presented in such a way that it is easy to understand and relay to responding officers. The data is stored in a way that facilitates dissemination to investigators already in the

field. According to the City of New York, the goal of the RTCC "is to give police officers and detectives current, comprehensive information in order to improve policing" (New York City Global Partners' Innovation Exchange, 2006, Goals and Objectives, para 1). ILP and COMPSTAT can be used to explain how law enforcement agencies deal with and respond to crime, but they do not explain the internal or external factors associated with why law enforcement agencies undergo change.

Since law enforcement agencies do not operate in a vacuum, the environment that they operate in often affect the decisions their leaders make. When presented with internal or external changes in their environment, law enforcement agencies can either simply cease activities and choose not to deal with the changing environment or adapt to the changes in the environment. Given their role in society, law enforcement agencies cannot simply cease operating and are instead better positioned to adapt to the changes. Therefore, organizational theories dealing with adaptational change, such as contingency theory, resource dependency theory, and institutional theory, are better suited to explain the internal and external pressures that cause law enforcement agencies to undergo change.

Historically, law enforcement agencies have been resistant to change, but given the state of policing in the United States today, they have almost been forced to change in order to adapt to the internal and external pressures placed on them (Ratcliffe, 2008). When examining an agency that created an RTCC partly due to external pressures, institutional theory can be applied to explain why an agency may choose to establish an RTCC. Technology plays a vital role in RTCCs, and the search to acquire or partner with organizations to obtain the necessary technology reveals that resource dependency theory may be useful in determining why agencies

are creating RTCCs. Further, in the examination of changes due to environmental factors, contingency theory may be beneficial.

By selecting organizational theories that offer the flexibility to explore why agencies are or are not implementing RTCCs, researchers have the best opportunity to examine what internal or external factors are influencing an organization's decision when it comes to the implementation of RTCCs. The following will be a review of contingency theory, resource dependency theory, interorganizational relationships (IORs), and institutional theory. Table 1 in Appendix A provides a summary of each of the key elements of the organizational theories discussed in this study.

Contingency Theory

Contingency theory can help explain why agencies choose to establish RTCCs. This is a prominent organizational theory that is often used to study organizational structures in law enforcement. Contingency theory states that law enforcement agencies must continually change for them to be effective. Agencies change when the environment in which they operate changes in some way, and they must change to adjust to the new environment (Pennings, 1987).

Contingency theorists argue that organizations must adapt to their environment to achieve high performance, thus ensuring their survival (Donaldson, 2001; Woodward et al., 1965). Contingency theory consists of three core elements: the contingencies; the organization's structure; and a proper fit between the organizational structure and the contingencies (Donaldson, 2001). Contingencies include the environment (Burns & Stalker, 1961), the size of the organization (Child, 1975), and the organization's strategy (Chandler, 1962). Having the proper fit between the organizational structure and the contingencies ensures the organization stays fit, which subsequently leads to high performance (Van de Ven & Drazin, 1985).

When an organization decides to change, that decision may be based on internal or external factors or a combination of both. Therefore, any effective organizational change requires fit to the contingency factors present in the environment (Donaldson, 2001; Van de Ven & Drazin, 1985; Pennings, 1987). Failing to account for the environmental contingencies may cause organizational change to lead to programs that do not get fully implemented or cause the organization to operate at low performance (Hollenbeck et al., 2002). If an agency in a law enforcement setting does not evolve according to changes in the environment, it may be seen as not being effective.

Not adapting to such environmental contingencies can cause a law enforcement agency to fall out of fit, which could in turn hurt the department's overall effectiveness. Three types of environmental factors have been identified in the literature on contingency theory. The first is the adaptation to the task environment, the second is the adaptation to new technology, and the third is the adaptation to the scale of production (Zhao et al., 2010).

These adaptations described by contingency theory can be applied to agencies that have decided to establish RTCCs. Contingency theory would suggest that agencies within a certain environment, such as a heavily populated, urban downtown area with sufficient infrastructure to support a robust camera system, would be more inclined to establish an RTCC. Likewise, a larger agency with access to greater resources and a bigger budget would be more likely to set up an RTCC. Finally, an agency that is forward-thinking, used to adapting to change, and more readily available to embrace technology would be more likely to establish an RTCC. Using contingency theory to examine an organization's decisions to establish an RTCC allows the primary investigator to explore the internal and external factors that may have led the organization to believe they had fallen out of fit and why the establishment of an RTCC was needed to restore fit.

Resource Dependency Theory

While contingency theory explores the idea of fit and organizational adaptation to regain fit, resource dependency theory deals with the interaction between agencies to acquire resources. Resource dependency theory helps explain whether RTCCs stand alone or whether they interact with organizations. This theory views organizations as an open system, where contingencies in the external environment affect the way in which they operate (Pfeffer & Salancik, 1978). When examining organizations, resource dependency theorists recognize that internal forces and the values and beliefs of the organization's leadership matter, but the decisions made at an organizational level are often based on external forces present in the organizational environment (Pfeffer & Salancik, 2003).

Resource dependency theory states that if an organization wants to survive, it must obtain the necessary resources needed to be seen as legitimate. Ensuring this survival often means partnering with another organization to obtain the necessary resources (Pfeffer & Salancik, 2003). Such partnership between two organizations may cause one organization to change certain aspects of its features, such as its structure or leadership, to continue to acquire resources essential for their survival (Pfeffer & Salancik, 2003). Since resources are being consumed by the organization, society is constantly evaluating if the organization's outputs are worthy of the resources being utilized (Parsons, 1956). As such, Parsons argued that legitimacy is essential for an organization's survival. The legitimacy need not be unanimous across the entire citizenry; however, it is required across a broad coalition of society members who contribute the necessary resources and benefit from the outputs (Parsons, 1956).

Since an organization is part of a larger social system and obtain necessary support from that system for survival, the organization's goals, activities, and outputs must be acceptable to

the larger social system (Pfeffer & Salancik, 2003). Maurer (1971, p. 361) defines legitimation as "the process whereby an organization justifies to a peer or superordinate system its right to exist, that is, to continue to import, transform, and export energy, materials, or information." It is important to note that individuals outside the organization bestow legitimacy upon the organization. Therefore, it is crucial that these individuals interpret the actions of the organization and deem them within the norms of social values (Pfeffer & Salancik, 2003). Since individuals who are not a part of the organization determine its legitimacy, the organization's leadership serves an important role in assuring that the organizational environment is in line with the social norms and values, thus taking active measures to ensure social legitimacy (Pfeffer & Salancik, 2003). These measures ensure that social norms or environmental changes are dealt with accordingly before the organization loses any legitimacy. The internal and external factors that organizations evaluate in their quest for resources to be seen as legitimate can be applied to law enforcement agencies that have or have not chosen to establish RTCCs.

With regard to RTCCs, resource dependency theory would suggest that law enforcement agencies that are not seen as legitimate may partner with other agencies that have already established RTCCs to be perceived as more legitimate. With technology playing such a large role in RTCCs, resource dependency can be used to explain any external changes an agency undergoes to be seen as more legitimate to outside entities. These changes may become vital as agencies seek to increase access to technology and data. A law enforcement agency will want to partner with as many outside entities as possible to have as much access to data as possible, and any perception of a law enforcement agency being not legitimate may affect its access to data.

Interorganizational Relationships

Given how important technology and data are to the formulation of an RTCC and that much of the data is controlled by entities outside law enforcement organizations, it is essential to understand the reason why agencies may decide to form an interorganizational relationship (IOR) with these companies. IORs deal with the relationship between and among organizations. IORs between law enforcement agencies and organizations can be vast. Law enforcement agencies form IORs with businesses to provide goods and services that they need to function as such. Popular goods and services include weapons, vehicles, technology, training, canines, aviation and marine assets, and body armor. Law enforcement agencies also enter into IORs with other law enforcement agencies. For example, a law enforcement agency may form an IOR with a neighboring law enforcement agency that allows each to enter one another's jurisdiction without seeking prior approval. This is called mutual aid. Law enforcement agencies can also form IORs to enhance their RTCCs.

For an RTCC to be effective at leveraging the vast amount of data in their possession in a relatively short period and then combine the data with a multiple of sensor data, law enforcement agencies need technology. One example of such technology includes programs law enforcement agencies have created to allow private businesses to install cameras that are in turn monitored by the RTCC. Private businesses provide video feeds to the RTCC, and these video feeds are seamlessly integrated into the RTCC. In these scenarios, both entities reap benefits. The businesses have their cameras monitored by a law enforcement agency, and the agency has immediate access to video feeds should a crime occur within the cameras' field of view.

Why a law enforcement agency chooses to enter into an IOR has been broken down into six reasons. These reasons include necessity, asymmetry, reciprocity, efficiency, stability, and legitimacy (Oliver, 1990). Oliver classifies these six reasons into two distinct categories: those that are voluntary and those that are non-voluntary. The non-voluntary reason is termed *necessity* and usually involves an organization being mandated by another such as a governmental regulatory agency to enter an IOR (Oliver, 1990). The five other reasons, namely asymmetry, reciprocity, efficiency, stability, and legitimacy, focus on voluntary reasons as to why an organization may enter an IOR.

Asymmetry

If an organizational enters an IOR for self-interest, it is doing so because of asymmetry (Oliver, 1990). An organization's desire to control and obtain scarce resources may cause it to try to limit its competitors' access to those resources, such as in the case of limited grant funds or access to other federal programs (Brewer et al., 2007). Organizations may also attempt to acquire knowledge limited to specialized law enforcement units. Popular examples include money laundering expertise, sophisticated surveillance techniques, and wiretap experience. Once the agency no longer participates in that IOR, this knowledge is transferred to the home agency (Hayeslip & Russell-Einhorn et al., 2002). Asymmetry has been associated with resource dependency theory.

Reciprocity

When entering an IOR based on reciprocity, organizations are doing so because the relationship would be mutually beneficial to both because of common goals or interests (Oliver,

1990). When asymmetry cannot explain why an organization entered an IOR, reciprocity can often be used to explain the decision (Oliver, 1990).

Efficiency

Efficiency deals with internal, rather than external, reasons for engaging in an IOR and is prompted by an organization's attempt to improve processes that lead to a better input/output ratio (Oliver, 1990). The need to be more efficient is based on internal workings and does not focus on the control of resources, exerting power over another organization, or playing well with other organizations (Oliver, 1990). If the costs associated with the IOR are too high or not worth continuing, the organization can choose to continue obtaining that resource by moving the production in-house or seeking it from another source at an acceptable price (Scott, 2014). Police reform is often centered on improving efficiency in some aspects of organizational processes and practices rather than making wholesale changes to how the police operates (Bullock, 2013).

Stability

When organizations experience uncertainty, they can choose to enter IORs to bring stability and predictability to their situation (Oliver, 1990). When an organization enters an IOR to gain stability, it is using coping strategies to "forestall, forecast, or absorb uncertainty" (Oliver, 1990, p. 246) to bring order and certainty to resources (Benson, 1975). While not salient in the United States, stability has been extensively researched in Central and South America. Studies indicate that when political stability is high, law enforcement and the military have a favorable view of the citizenry. Such favorable view decreases the need for law enforcement to partner with its military

counterparts in anticipation of civil unrest (Sung et al., 2022). Stability has been associated with resource dependency theory.

Legitimacy

Based on institutional theory, legitimacy attempts to explain IORs through a lens that organizations wish to abide by a set rules or beliefs that have been established by partners outside the organizations (Oliver, 1990). This outward search for validation of the organization means that one organization may be seen as more legitimate than another. Hence, organizations that form relationships partly due to legitimacy will often do so with an organization that appears more legitimate than themselves (Oliver, 1990). Often, factors such as organizational age influence the perceived legitimacy of an organization, and young organizations may partner with established organizations to quickly establish legitimacy (Wiewel & Hunter, 1985).

While the reasons organizations establish IORs are not coupled with a specific theory, they can be used as an enhancement to further explain organizational decisions. Many of the reasons why organizations choose to enter an IOR are based on popular organizational theories. Referencing these reasons along with the derived organizational theory serves to provide further background to internal and external factors that might explain why organizations choose to engage in certain behavior.

Institutional Theory

One such organizational theory and the closely associated concept of legitimacy is institutional theory. Institutional theory helps explain the decision law enforcement agencies make to establish an RTCC. This theory has been touted as one of the organizational theories

that provides the best framework for explaining why law enforcement agencies structure themselves the way they do and engage in the types of activities they do (Giblin & Burrus, 2008). While resource dependency theory and institutional theory are somewhat similar in that outside forces play a powerful role in organizational change, institutional theorists hypothesize two distinct lines where institutional theory applies to policing. The first line focuses on "myths" and the institutional environment rather than the demands of work (Meyer & Rowan, 1977). The second one centers on why institutions change and become like other organizations; this phenomenon is called institutional isomorphism (DiMaggio & Powell, 1991).

To gain legitimacy, organizations must structure themselves and operate in a way that is amenable to their stakeholders (Giblin & Burrus, 2008; Meyer & Rowan, 1977). Organizations require a wide variety of materials, resources, and technical information if they are going to survive. Besides these material resources, organizations must be accepted and seen as creditable to maintain success over a long period of time; in essence, they must be perceived as legitimate (Scott, 2014). Suchman defined legitimacy as "a generalized perception or assumption that the actions of an entity are desirable, proper, or appropriate within some socially constructed system of norms, values, beliefs and definitions." Legitimacy is applied generally to an organization rather than an event-specific evaluation and is "possessed objectively yet created subjectively" (Suchman, 1995, p. 574).

This willingness to appease the stakeholders means the organizations' primary goal is to acquire legitimacy and operate in the most efficient way (Meyer & Rowan, 1977). The environment in which an organization operates can consist of individual institutional sectors that can affect the organization (Meyer & Scott, 1983). These institutional sectors may comprise

community leaders, elected officials, governmental regulators, accrediting bodies, and professional unions (Matusiak, 2018).

Through repeated interactions, an organization can garner legitimacy with these institutional sectors, and it eventually begins to become independent of the environment (Aldrich, 1999; Donaldson, 2001; Maguire, 2014). The more the organizations become independent of the environment, the more they are allowed to decide how to structure themselves and what the best strategies are, even if it is only symbolic (Maguire, 2014). As a result of legitimacy, organizations are under a tremendous amount of pressure to act in a certain way. The act of appearing legitimate appears to be primary, and whether the pressure makes the agency more efficient or effective seems secondary to appearing legitimate (Meyer & Rowan, 1977).

As agencies begin to act in a way that is expected of them, they begin to acquire legitimacy, and the more legitimacy the organizations achieve, the less likely they are to have their activities questioned (Meyer & Rowan, 1977). Agencies need to be seen as legitimate because this ensures a constant flow of resources necessary to perform their operations (Greenwood & Hinings, 1988).

When an agency does not respond to institutional pressures calling for changes, it runs the risk of losing legitimacy and potentially losing resources, which can ultimately lead to the organization's failure (King, 2000).

There are only a certain number of ways an agency can change to maintain its perceived. Because of this constraint, similar organizations begin to look alike. DiMaggio and Powell called this phenomenon "institutional isomorphism." Isomorphism is a "constraining process that forces one unit in a population to resemble other units that face the same set of environmental conditions" (DiMaggio & Powell, 1991, p. 149). DiMaggio and Powell (1991) describe three

processes that lead to isomorphism: coercive, mimetic, and normative processes. Organizations undergo coercive isomorphism when forced to do so by another powerful organization, such as a regulatory or administrative body (DiMaggio & Powell, 1991). The next source of isomorphism is the mimetic process. When organizations copy what appears to be a successful process, they engage in mimetic isomorphism (DiMaggio & Powell, 1991). The final source that leads to isomorphism is the normative process. As information, education, and training are disseminated in a particular discipline through professional associations or training conferences, attendees will return to their agencies and begin to implement what they have learned. This process of implementing what has been learned in these settings is referred to as normative isomorphism (Roy & Seguin, 2000).

Studies have revealed that there are institutional pressures on police agencies, and these pressures can be responsible for a law enforcement agency moving in a specific direction. Examples include the study by Katz et al. (2001) that showed that at least one police department created a gang unit even though a gang problem did not exist in its jurisdiction. Willis et al. (2007) found that strong institutional pressure to appear progressive and modern was the primary factor for the adoption of COMPSTAT in the three law enforcement agencies studied. Further, Carter (2016) revealed that strong institutional pressures following the September 11 terrorist attacks played a significant role in police decisions to adopt ILP.

Institutional theory is important to this research because numerous studies have found that there are institutional pressures on law enforcement agencies and that these pressures have caused law enforcement agencies to adopt programs or policies when there was not any need or necessity for them.

The internal and external pressures described in institutional theory can be applied to a law enforcement agency's decision to establish an RTCC. Such pressures could, for instance, come from elected officials, or city leaders, pressuring agencies to address a rising crime problem. The agency may establish an RTCC not because it is the best solution that fits their needs but because they are trying to gain legitimacy. Additionally, if a law enforcement agency establishes an RTCC simply because a neighboring agency did the same, the institutional external pressures of isomorphism could be used to explain this decision. By using institutional theory in this study, it is believed that the theory can be applied to these sorts of internal and external pressures that could cause a law enforcement agency to set up an RTCC in an attempt to appear more legitimate.

Summary

In summary, the literature regarding a law enforcement agency's decision to establish an RTCC is lacking. On the surface, RTCCs seem to incorporate elements of reactive and proactive forms of policing, yet these forms of proactive policing could not fully explain why law enforcement agencies establish RTCCs. Adaptive theories of organizational change were also utilized to explain whether internal or external pressures play a role in the agency's decision-making process. By employing the organizational theories discussed above, the researcher has the best opportunity to examine what internal or external forces are influencing law enforcement agencies' decisions to implement RTCCs. Finally, these theories allow researchers to design studies and instruments that capture whether these internal or external pressures are considered when an agency decides to establish an RTCC.

CHAPTER 3: METHODS AND DATA

While organizational theories explaining organizational change and theoretical frameworks explaining IORs are well-researched, the application to RTCCs is in the early stages. As such, this study focuses on the reasons why Florida law enforcement agencies decided to establish RTCCs. To do this, the study uses an exploratory research design that focuses on why Florida law enforcement agencies are establishing RTCCs. Hopefully, this exploratory research will lead to more research in the future.

Research Design

The overall study used a mixed methods design to examine an organization's reason(s) for or against creating or implementing an RTCC. The study was broken up into two different studies (Study 1 & Study 2). The research design for Study 1 consisted of a content analysis of newspaper articles. The research design for Study 2 included interviews utilizing open-ended questions posed to Florida law enforcement agencies' command staff with knowledge of why their agency has or has not decided to establish an RTCC.

This methodology is appropriate for two reasons. First, research into why law enforcement agencies are creating RTCCs is non-existent. A content analysis of newspaper articles will provide a baseline for establishing what newspaper writers, the public, and law enforcement agency administrators publicly express as important with respect to RTCCs. However, a content analysis will not provide the critical information related to specific reasons as to why a law enforcement agency has or has not created an RTCC. Structured interviews of Florida law enforcement agencies' command staff with knowledge of how the decision to create an RTCC was reached will provide the most reliable information on agencies that have developed an RTCC. Commonalities

and differences among the responses will allow the development of an overall explanatory framework that can explain why this approach is spreading. Appropriate qualitative analyses will be used for each type of data collected.

Study 1

For the newspaper content analysis portion of this study, inductive reasoning with conventional content analysis, as outlined in Hsieh and Shannon (2005), is used. Inductive reasoning using conventional analysis is appropriate because the data will allow the researcher to develop the important themes contained in the articles. Conventional content analysis is appropriate when the literature on a given subject is limited (Hsieh & Shannon, 2005), and researchers avoid using preconceived categories and allow the data to establish the coding categories (Kondracki et al., 2002). For this study, the unit of analysis is words or phrases.

Process

Based on the work of Corbin and Strauss (2015) establishing sources of theoretical sensitivity, a literature review of peer-reviewed articles on fusion centers was conducted. Several broad categories emerged from these articles, augmented by the primary researcher's professional experience in working in and around fusion centers. These broad categories include the following:

- Technology
- Organization
- Personnel
- Structure
- Marketing

Once the broad categories were identified, a spreadsheet was created that contained columns to capture whether any of these categories were observed in the articles.

This study used a purposive, non-probability sampling method to obtain news articles for analysis. Purposive sampling is the most common sampling form used in content analysis and ensures that the primary researcher obtains suitable informants that contain the best subject-matter knowledge associated with the research topic (Kyngas et al., 2011).

To obtain the appropriate news articles, Nexis Uni was used to search for all newspaper articles in the collection that contained the term “real-time crime center” or “real time crime center”. Nexis Uni captures 21 publication types, so under the “Publication Type,” only “Newspapers” were selected. This filter was the only filter used for the dataset. The total dataset for Study 1 consisted of 235 news articles from all over the United States as well as international newspapers. All results listed in the search are made part of the study’s content analysis dataset.

As articles were being read, specific themes within these broad categories were added to the spreadsheet. If a column was inserted to the spreadsheet, then all the previously reviewed articles were reviewed again to ascertain if the specific category for the new column was present in any of the already reviewed articles. This process was continued until the entire dataset was read once.

The articles were read a second time to determine general themes that were not captured in the first reading. In the second reading, concepts that were not associated with the literature review or professional experience were focused. If a concept or theme emerged that was not developed from the literature review or professional experience, a column was added to the spreadsheet. The column was distinguished in such a way as to notate that it was from the second reading as opposed to the first. As already mentioned, if a column was added to the spreadsheet,

then all previously reviewed articles were reviewed again to ensure that the concept associated with the column is not present in previously reviewed articles.

Once all articles were reviewed a second time, the concepts or words entered in the spreadsheet were reviewed to see if any of the concepts or themes could be condensed or shortened into specific themes or phrases. Thereafter, a third reading was conducted to officially code the articles using the codes created during the first two readings. The codes were entered in the spreadsheet and marked as codes from the third reading. Coding was developed for the existence of a concept and the frequency of a concept. Both terms are important as the existence of a concept may be a reason for the creation of an RTCC, and the frequency of a concept may signal to the researcher that a concept is an important aspect of RTCC and should be covered in detail during the interviews.

Once all the coding had been completed, the results were entered into Statistical Package for the Social Sciences (SPSS) for analysis, and the frequency for each code was obtained. After obtaining the frequencies for each code, the results were used as a guide of important concepts and themes during the interviews.

Initial data

From the first reading of the news articles, it became apparent that cameras are vital to an RTCC as 105 of the 235 (45%) news articles mentioned cameras. Privacy concern was the second most discussed item in the news articles (45 articles, 19%) mentioning privacy concerns associated with RTCCs. Other notable topics included license plate readers (22 articles, 9%), ShotSpotter technology (20 articles, 8%), and facial recognition technology (14 articles, 5%).

Analysis

From the second reading, three concepts emerged. The first is that RTCCs facilitate law enforcement agencies transitioning from a strictly reactive model of policing to a sort of enhancement of reactive policing. An enhancement to a reactive form of policing is appropriate as opposed to calling it proactive policing because the crime is already afoot, and law enforcement is using technology to detect it at much higher rates than manually. With the speed and efficiency an RTCC can process data and catch crime as it occurs, a new model of policing might need to be created. Such a model needs to bridge the gap between having citizens determine when to involve the police and preventing crime before it occurs; a model that is created based on the newfound efficiency with which law enforcement can now operate.

A new model might be necessary because this "catch and investigate crime as it occurs" model does not fit the reactive and proactive model of policing. Reactive policing involves three main functions of law enforcement: routine patrol, immediate response to calls, and follow-up investigations (Cordner & Sheehan, 1999). RTCCs appear to enhance reactive policing by generating the calls themselves based on the license plate reader (LPR) hits for stolen vehicles, informing patrol officers of ShotSpotter notifications, and using cameras to solve crimes that in the past would not have been possible. This concept does not fit the reactive model of policing because in reactive policing, the public decides whether to involve the police, and the RTCC keeps the involvement of the police within the police agency itself, much like in the proactive model (Waddington, 1993).

Proactive policing can be defined as "all policing strategies that have as one of their goals the prevention or reduction of crime and disorder and that are not reactive in terms of focusing primarily on uncovering ongoing crime or on investigating or responding to crimes once they

have occurred" (National Academy of Sciences, 2018, pp. 1, 30). RTCCs do have a proactive component, namely CCTV cameras. However, the concept of law enforcement agencies using CCTV cameras is not a novel one. The novel concept appears to be the use of technology and staff to monitor the cameras and marry the camera feeds with police calls for service along with other law enforcement databases RTCC employees have access to. Once RTCCs are made aware of a crime, it is no longer a proactive incident; the crime has already been committed, so the proactive model of policing no longer applies.

RTCCs seem to facilitate this new model of policing but also allow law enforcement agencies the flexibility to provide a vast amount of intelligence, including surveillance camera footages, to investigators within a short time frame. This analytical component to RTCCs is highlighted in the content analysis as eight articles discussed the organization of the agency's RTCC. In all cases mentioned, the RTCC was in the agencies' intelligence unit. Placing the RTCC within the Intelligence Bureau means that some level of intelligence analysis is taking place and the agencies that have established an RTCC are now dependent on a crime intelligence analyst. However, the content analysis revealed that sworn (both officers and detectives) and non-sworn personnel can be assigned to an RTCC. This organizational change or reorganization to accommodate a structural change aligns with idea that an organization will seek to attain fit. The idea that agencies will seek to obtain fit is a central tenet of contingency theory.

The second concept is that RTCCs require a tremendous amount of electronic equipment and software. Numerous articles discuss having data warehouses and specialized software capable of searching, retrieving, and disseminating large volumes of data to investigators in the field. As agencies determine they are no longer in fit with the technology requirements they need

to properly support the RTCC, they may consider upgrading the technology that supports the RTCC. This idea of regaining fit is associated with contingency theory.

Agencies that establish RTCCs will need to rely on outside software vendors to supply, maintain, and update the necessary software and computer hardware necessary to support the technological backbone present in an RTCC. Given the rate at which agencies are establishing RTCCs, it does not appear that the technological resources required to establish an RTCC constitute a factor. However, how these agencies dealt with these issues does warrant examination for agencies that could be considering an RTCC.

The final concept emerging from the second reading is that RTCCs act as a force multiplier for law enforcement agencies that have many vacancies. For this study, the term "force multiplier" will be used in the context where RTCCs may substitute or augment for a lack of sworn officers. Three articles discussed the need to address department-wide vacancies and RTCCs appear to offer a solution that allows law enforcement administrators to fill in gaps in departmental staff. As agencies face an increase in crime, leading to extended wait times for an officer to respond, they may begin to experience external pressures from a variety of sources. The use of an RTCC as a force multiplier to appease the entities exerting the external pressure can be associated with institutional theory.

While not specifically noted in these three concepts, resource dependency theory was observed during the content analysis. At least one city devised a program that formed a partnership between private businesses and the RTCC. In this program, private businesses purchased a surveillance camera compatible with the RTCC's camera system. The business paid for the camera and its installation, while the RTCC paid for the video storage. The business had their video camera monitored 24 hours a day by the RTCC, and the RTCC had instantaneous

access to the stored camera video footages in case crimes occurred near the cameras. This public–private partnership where both parties obtain benefits represents a central tenet of resource dependency theory and the tenant of reciprocity and efficiency of Oliver (1990). The program was so successful that at least one article discussed the possible expansion of the program to residences located in high-crime areas.

The results of Study 1 indicate that organizational aspects, electronic equipment, and staffing issues were salient regarding the development of RTCCs. To further explore these aspects, specific questions were tailored to discuss the points identified in Study 1. The organizational aspect was explored with questions that served to ascertain the agencies' intelligence structure and current capabilities. Questions were also formulated to investigate how the organizational structure changed because of establishing an RTCC. Technology issues were examined within the context of funding with several questions devoted to grants. Grant questions included whether grants were used, who wrote them, and whether the agency partnered with academic institutions. Finally, staffing issues were addressed through questions eliciting information on who was assigned to work in RTCCs and the allocation of resources to the various units within the law enforcement agencies. These points along with their corresponding questions were examined in Study 2 via the adaptive theories of organizational changes presented in this thesis.

Study 2

In qualitative research, data practices and collection procedures are the main factors that influence the quality and trustworthiness of a research project, with interviews being the most used form of data collection (Kitto et al., 2008; Taylor, 2008). Within the subset of interviews, semi-structured interviews are the most used interview technique in qualitative research

(DiCicco-Bloom & Crabtree, 2006). Semi-structured interviews have been described as in-depth interviews that allow respondents to answer formulated open-ended questions that are answered conversationally and often accompanied by follow-up questions involving why or how (Adams 2015; Jamshed, 2014). The format of the semi-structured interview gives the researcher the flexibility to navigate through the topics on the survey questionnaire and delve into totally unforeseen issues, unlike a standardized survey that requires strict adherence to the questionnaire (Adams, 2015).

The semi-structured interview format is well-suited for several purposes including interviews that require follow-up questions, probing open-ended questions that allow the researcher to learn about the independent thoughts of each interviewee, or when a researcher ventures into uncharted territory and requires freedom to spot emerging trends and wishes to pursue them (Adams, 2015). Individuals who are well-suited for semi-structured interviews include administrators, front-line staff, top administrators, or board members (Adams, 2015). The semi-structured interview format is appropriate for this study because the study purpose is to scrutinize law enforcement agency administrators' perceptions of or opinions on a complex issue, and semi-structured interviews allow participants to express diverse perceptions of a particular topic (Barriball & While, 1994; Cridland et al., 2015).

Process:

For Study 2 Florida law enforcement agencies that had established RTCCs were readily available through internet searches. Additionally, through law enforcement associations, a list of Florida law enforcement agencies that had established RTCCs were identified. To minimize the non-response rates, a personalized letter introducing the primary investigators was included in the packet. The letters were personalized for each recipient, contained blue ink signatures, and

presented on high-quality paper. The letter requested an appointment for an interview of approximately one hour with the sheriff or police chief, or an individual with knowledge regarding their agency's decision to create an RTCC. The letter explained the purpose for conducting the research study, namely to explore the reasons why Florida law enforcement agencies have or have not established RTCCs. The letter asked for their help in conducting the study and notified them that their responses to the questionnaires will be confidential.

For this study the semi-structured interview guide formulated by Kallio et al. (2016) was utilized. The interview guide consisted of five phases that included (1) identifying the need for using semi-structured interviews; (2) retrieving and using existing knowledge; (3) formulating a preliminary interview guide; (4) testing the guide; and (5) presenting the completed guide (Kallio et. al., 2016).

Instrument:

For this study, phase 1 and phase 2 had already been completed, so the guide begun with phase 3. The interview instrument used key points

developed in Study 1, key points associated with each organizational theory and the primary investigators knowledge of RTCCs collected or observed through experiences in the law enforcement community. The instrument consists of 18 questions, with several questions containing multiple parts and follow-up questions.

For phase 4, the primary researcher conducted a mock interview of a law enforcement official who was familiar with RTCCs, but whose agency did not have an RTCC. This agency was also not one of the Sheriff's Offices that did not have a RTCC but was provided a packet. Questions were reviewed to make sure the target of the study stayed at the organizational level

and no identifying information was collected that would identify study participants. The instrument is attached as Appendix B.

For phase 5, the instrument would be provided to potential agencies with the introductory letter, so agency administrators could review the questions beforehand and decide whether there were sensitive questions they did not feel comfortable answering. The packets advised participants of the goals and objectives of the study in accordance with the guidelines set by the University of Central Florida Institutional Review Board (IRB). A copy of the UCF-IRB approval letter is listed in Appendix C.

In total 53 packets were mailed to various Florida law enforcement agencies. The agencies that currently have RTCCs consisted of 18 sheriff's offices, 11 police departments and 1 University Police Department. To research the possibility of why certain sheriff's offices had not established RTCCs, the 2020 census data regarding county population data were collected from the United States Census website. The population of each of Florida's 67 counties was obtained and the county with the lowest population that currently had an RTCC was determined. That county is currently Flagler County with a population of 115,378. Flagler County established the minimum county population for including a sheriff's office that currently did not have an RTCC.

The counties were then sorted from most populous to least populous and the 20 counties with the highest populations that currently do not have an RTCC were included in the list to receive packets. All the sheriff's offices that received packets and that currently do not have an RTCC had higher county populations than Flagler County. Since a major Florida university

partnered with other law enforcement agencies in the creation of an RTCC, 3 large state universities with police departments also received packets.

To collect as much data as possible, 30 days after the first packet was mailed, a second packet was mailed to the 17 agencies that currently have RTCCs but did not respond to the original mailing. The second mailing contained many of the same components as the first mailing, but the personalized letter was different. The personalized letter in the second mailing listed the number of agencies that had already agreed to participate in the study and emphasized the researchers desire to obtain as much data as possible. See Table 2 in Appendix A for a breakdown of the types of law enforcement agency that were sent an instrument, whether the law enforcement agency had an RTCC and the agency's response rate.

Population and Sample:

The subjects of this study represent those law enforcement agencies that responded to the mailed packets and agreed to participate in an interview or completed the instrument and returned it to the primary investigator. After the first mailing, 20 of the 53 agencies that were sent a letter responded, representing a response rate of 37%. Of these 20 agencies, one agency advised they would not participate in the study and 7 agencies advised they would participate in the study but failed to schedule an interview. The remaining 12 agencies provided data for this study. Ten agencies chose to participate in an interview and 2 agencies chose not to participate in an interview but completed the instrument and provided the results to the primary investigator. Of the 12 agencies that provided data, 10 currently have RTCCs and two do not. The 12 agencies consisted of 8 sheriff's offices and 4 police departments.

After the second mailing, 1 additional sheriff's office responded. This sheriff's office had an RTCC and chose to provide data by completing the instrument and returning it to the primary researcher. This brought the final total number of agencies that provided data for this study to 13.

Analysis of Data:

A content analysis of the data collected during the interviews was conducted and 9 interviews were recorded and transcribed. One agency that participated in an interview requested that the interview not be recorded. For this interview, the primary researcher took notes and typed key responses in a document. For the three agencies that provided data via the instrument form, the entire form was loaded into the analysis software. The content analysis was conducted on Nvivo version 12.

Findings

The purpose of this study was to conduct an exploratory analysis to see if Florida law enforcement agencies are creating RTCCs to respond to and investigate crime in "real time." Additionally, the following questions were asked: Did external factors (e.g., elected officials, clergy, judges, community leaders, etc.) play a role in the Florida law enforcement agencies' decision to establish an RTCC? Did internal factors play a role in the Florida law enforcement agencies' decision to establish an RTCC? Do RTCCs stand alone or do they interact with other organizations?

The answers provided by respondents during the interviews are categorized and summarized as follows:

Question 1: Is your agency an intelligence-led policing agency?

Nine of the 13 agencies (69%) stated their agency was an intelligence-led policing agency. Many of the respondents mentioned having weekly intelligence-sharing meetings where analysts would brief agency command staff on crime trends, including hotspots and crimes rates. A central theme was the use of data to determine the deployment of resources. One respondent indicated as follows:

> Yes, when I came into office, that was one of my goals, to use an intelligence-led policing approach. And we've used the Omega Dashboard for several years to help us visualize our data on crime, and every Monday at our command staff meeting, we have the operational division commanders stay after command staff, and we go through what our crime issues are from the week before. We look at, you know, the number of calls for service, number of burglaries, where we're felons have registered, domestic violence calls, just basically a whole a lot of data. It's probably a thirty-something page report that we go through. We've got our county broken down, of course, into zones, but we also create an east and west district. So that's kind of how we look at our data.

One agency also described how it creates intelligence bulletins of suspects and suspects' vehicles, which are passed on directly to patrol and allow patrol officers to prioritize their time during their shift. One respondent said the following:

> When I got hired here in 2004, I was the very first analyst that was hired. So, when I started, I had to build everything. So, we had an intel function, but we didn't have an analytical function. So, from there, fast forward to today, you know, we do products daily that patrol uses to make their plans for the day, the week, the month, whatever.

The three agencies that reported they were not intelligence-led policing agencies seemed to strictly abide by the academic definition of intelligence-led policing. One agency relayed that the answer to this question was yes and no. The respondent mentioned as follows:

> We follow parts of intelligence-led policing, but not all of them. We do work proactively to identify those offenders that are causing the most crime. Once those offenders are identified, we work with our proactive units to target them. However, we don't focus on the criminal organizations that are committing crime, and for that reason, I would say that we are not an intelligence-led policing agency.

Another respondent said the following:

> I think we utilize some components of intelligence-led policing. I think we're probably more accurately defined as a problem-orientated, problem-solving agency. I would say the strategy that we articulate to our body is what we refer to as precision policing. You know, it's, I don't know if it's necessarily as mainstream as some of the other ones. It's not uncommon, though. It's kind of was more of a stolen strategy from New York, where we really tried to focus on chronic offenders. So, like intelligence-led policing, we're looking at data that implies what is going to be reoccurring problems, reoccurring people, reoccurring issues, and kind of just zero in and focusing on problem-solving those issues.

Question 6: Was the primary reason the RTCC was established a criminal event?

One law enforcement agency that does not have an RTCC but does have a center like an RTCC created its crime fighting center to deal with a crime hotspot. The high-crime area covered a three-square-mile area near a major state university. The center became so successful that the law enforcement agency decided to expand camera coverage, and it now has several hundred cameras in-high crime areas. As the center grew and expanded,

> it integrated additional technology, such as LPRs. The law enforcement agency does not consider their crime fighting center to be an RTCC because employees of the center do not assist with crimes in progress. The employees simply react to crimes the technology detects, and the employees alert the appropriate patrol officers.

Question 7: If the primary reason for the creation of the RTCC was something other than a criminal event, could you explain what was the reason?

Of the 11 agencies that responded that currently have an RTCC, 9 (81%) indicated that the availability of technology was a primary driver of the agency's decision to establish the RTCC. The level of technology that was discussed varied between agencies; at a basic level, it appeared to be availability of cameras and LPRs all the way to advanced technology that tied these technologies together with calls the departments had received. These advance forms of technology meant that the computer programs graphically displayed the location of cameras and license plate readers on a map and often prominently showed the relevant camera feed for the call the RTCC was working. This meant that the RTCC was able to operate much more efficiently despite having access to hundreds of cameras and large volumes of data. One respondent stated as follows:

> But the reason why I believe it started is not because of an event, but because of technology, because everything seemed to be pushing that way. You know, the involvement of cameras and the ability to bring in cameras along with ShotSpotter or LPR, those systems just working together now, even though we don't really get into NIBIN been up here, that's more with the crime analysis unit but because of the technology can link cases.

Mention was also made of a specific incident and the respondent began looking for technology to ensure that a similar incident would not occur within the agency's jurisdiction, specifically reported as follows:

> There was an event that was a Miami kidnapping of a kid. And the person fled the Miami area in a white van, and the white van was discovered in the county north of me in Pasco County. It amazed me, annoyed me. I couldn't believe that. A van could get all the way from South Florida up to Pasco County when all these Amber Alerts were all going all over the place and stuff. Then we didn't have one LPR hit or anything like that where we could have used it to kind of track the movements. So, I kind of looked at the technology that was available, and some of the technology that we had already gave us access to traffic cameras, and there was already red light cameras that have a component of catching tags.

The cost of the technology becoming cheaper was also listed as a reason why at least one agency created an RTCC. The respondent stated the following:

> Well, I've seen it, you know, successfully employed in other areas, in a primarily larger areas with more resources, where I would consider us a mid-size community. And this kind of, I'll be honest with you, the impetus of it is really technology disruptors in the industry that have come in and made it affordable for agencies our size.

The final reason given for the creation of an RTCC was mimetic isomorphism. A city police department had an RTCC, and the sheriff felt that it would not look well for a sheriff's office to not have an RTCC when a smaller police department already had one. The sheriff's office established an RTCC, but it was not supported. It was not until a new sheriff was elected

that the RTCC received the support of the agency's administration. The city later closed their RTCC and partnered with the sheriff's office.

Question 8: What are the primary functions of the RTCC?

All agencies with an RTCC that responded provided generally the same answer regarding the primary functions of their RTCC. That primary function involves providing situational awareness in real time to officers responding to a crime in progress. One respondent stated as follows:

> The real-time crime center (RTCC) vets calls for service to provide situational awareness and enhance officer safety, as well as processes investigative requests for homicides, robberies, shootings, and other critical event cases. The RTCC functions as a centralized data hub that rapidly mines information across multiple criminal, civil, and social media sources. The coordination and distribution of real-time information is supplied to police officers and detectives to provide situational awareness and ensure a more effective and timely response to criminal activity and subject apprehension while enhancing officer safety. Additionally, the RTCC is responsible for the immediate dissemination of ShotSpotter alerts. Once an alert is received, road officers are immediately advised of the location of the shots fired as well as the number of rounds that were fired.

Even though all agencies responded that the primary function of their RTCC is to provide information to responding officers, when the RTCC analysts close their call will depend on the agency.

At least 6 out of 13 agencies (46%) reported that when the call no longer deals with situations in real time, it is closed, and the intelligence developed during the call is packaged and sent to the crime analysis section. Three agencies out of 13 (23%) indicated that the analyst

could continue data mining for information regarding that call from previous calls if there were no pending calls. One agency stated that the line between the RTCC and the crime analysis section is blurry. Within this agency, often RTCC analysts will be allowed to continue to work on closed calls, but it depends on the type of case and their current workload. The decision for an RTCC analyst to continue working on cases no longer considered to be in real time is made by the agency's intelligence manager, who supervises the crime analysis section and RTCC. Several agencies mentioned that the support provided to the responding officer often depended on the experience of the RTCC analyst. One respondent described this interaction as follows:

> You get a call, review it, evaluate it, use your skills, your training and your experience to identify what's available that could support or help. Sometimes it's simply, as you heard, the officer's request; if they can get them a photo quickly of a particular person or something, they'll respond on the radio and have dialogue back and forth with the officers. Sure. You're going to get to the issues of the concerns or some of the problems that come up that will come back to that, but there is a dialogue between the real-time center and the officers in the field and pretty much a healthy thing for the most part.

For many of the agencies, this interaction between the RTCC and the responding officers will involve civilian analysts. Eight out of 13 agencies (61%) indicated that all employees that work in the RTCC are non-sworn analysts. This heavy emphasis on civilian analysts has created unique training opportunities for the analysts. Several agencies reported that they require the analyst to participate in police ride-alongs, attend training with the patrol officers, and even attend parts of the orientation for new officers. All this training is geared toward having the analyst think like an officer when they assist with a call and anticipate what the officer will need during that call and, more importantly, what that officer does not need to hear during the call.

The role of the analyst turned out to be pivotal to providing information to responding officers, yet in larger agencies, sworn law enforcement officers played a vital role in providing situation awareness to responding officers. One large police department used senior detectives in addition to analysts in its RTCC. The detectives operated the camera system, were responsible for communicating scene intelligence to responding officers, maintained and preserved video evidence, and proactively used the RTCC cameras to identify crimes in progress. The agency's representative acknowledged that most agencies that visited the RTCC were originally hesitant to remove detectives from case work, but once the visitors saw how the agency uses detectives in the RTCC, they were impressed at what the detectives from the RTCC could accomplish.

Question 9: How has the RTCC changed the way your agency operates?

Eight of the 11 agencies (72%) that established an RTCC indicated that the RTCC has made the police department more efficient, enabled the department to solve crime that would not have been solved before, increased officer safety, and made the public and suspects safer. One respondent stated as follows:

> I think it's just made us more efficient in our response. For the most part. I think there's a greater awareness of the officer in the field of what data is available to them, which I think makes them more efficient in their ability to close a case. I think it makes us safer in our ability to respond and not go in blindly. I certainly have visions of doing more with it.

Several agencies spoke about crimes that would not have been solved prior to the establishment of the RTCC. One respondent indicated the following:

> So the technology that we have, we just had a shooting that came in, the camera's kicked off in that area, the analysts were able to see the suspect vehicle that was shooting.

> They were able to see the targets who ran away. And it was not late at night, but it was at night. There was no one else on the street except for this one woman who was in between the shooter and the people they were shooting at. And she dove under a dumpster, so when the analysts saw that before the officers arriving, they're already telling the officers they're giving that overview of, hey, you may have somebody who's shot. Well, she gets up, she walks into a store across the street. The officers still have not arrived. Everybody's gone. That would have been a note in the log. But because of the ShotSpotter, and because of the real-time crime center, they were able to give a direction of vehicle description of the suspects. They track back with their cameras to where the suspects are. The victims had run to their car, and they drove off. They drove 15 minutes down the road and parked right under one of our other cameras, and the officers were able to be directed right to where they were. We were able to identify them as the victims in the shooting. The officers at the scene, were told that the woman had walked into the store. They pulled her out and just brought her out to the corner and went, "Is this her?" And the real-time crime center analyst was like, "Yep, that's her." So now because of the RTCC, we have suspects, we have witnesses, we have victims. The whole incident is on camera where a case is opened and closed, where it never would have been.

Finally, several agencies spoke about an increase in officer and public safety. One respondent narrated an incident where a deputy made a traffic stop, shortly after which, she communicated via radio that the suspect had attempted to grab her weapon. The RTCC was able to access the deputy's body-worn camera and could check on the status of the deputy. Several back-up deputies responded to the deputy's location at high speed with their lights and sirens on, many coming from several miles away. The RTCC was able to tell the responding deputies that

the situation was under control as the deputy was able to deploy her taser and place the subject under arrest. Using the video feed from the deputy's body-worn camera, the RTCC was able to confidently tell responding deputies that the situation was under control and an emergency response at high-speed rate was no longer needed. The respondent indicated that prior to the establishment of the RTCC, a coordinated response that kept the deputy and the public safe would not have been possible.

Question 10: What benefits have you seen because of the establishment of an RTCC?

Many of the respondents reiterated their answers to the previous question and discussed how the RTCC has fundamentally changed the way their agency operates. One respondent mentioned the following:

> So obviously, the benefit of being able to identify those suspects and get those suspects in custody. We had another incident where a guy was followed home from a bank and was murdered in his driveway. We were able to pick up with the LPR camera our victim, and right behind our victim is the suspect. Without that, we're never arresting that guy. How are we ever going to identify that guy? We know it's such a random thing. You're not getting that, so the real-time crime center and the assets that feed in there have certainly been beneficial in solving some very significant high-profile crimes.

Another benefit that one agency reported was having analysts from multiple police departments working together in the RTCC. The respondent stated as follows:

> You know, when you see multiple units within the agency working together, you know, or especially multiple agencies, you know, and that's probably been one of the biggest benefits to this crime center is with everyone's information coming through one hub. You know, now when we take a report in a city for a given crime and that has M.O. ties to

crimes that were committed in other parts of the county, well, we're all at the crime center. We get all those bulletins; we get all that information at one spot.

Question 11: What are some negative aspects of establishing an RTCC?

All 13 agencies that responded indicated the cost associated with instituting and maintaining an RTCC is the biggest negative. Agencies reported that the software used to aggregate disparate sources of information can cost over half a million dollars. Additionally, each camera and LPR system can cost tens of thousands of dollars to install and maintain. Finally, there are licensing fees for almost every database used in the RTCC. This means that selecting a product that does not work as intended or does not do as was promised means large sums of money are wasted. Several agencies discussed the pressure of ensuring that these sorts of mistakes do not happen multiple times.

Another negative aspect of establishing an RTCC is staffing. Two agencies indicated that having all the technology in the world is nice, but if agencies do not have qualified, competent analysts to utilize the databases and provide meaningful results, then the technology is essentially useless. In addition, analysts employed in an RTCC must be comfortable working under pressure to deliver accurate information, often within a few minutes. Often, analysts cannot adjust to the demands of working in an RTCC and are either dismissed or transferred out of the unit. This can be a significant loss to the agency that has invested time and resources training that analyst.

Question 13: Who in your agency was the primary person/role responsible for the implementation of the RTCC?

Seven of the 13 agencies (53%) that responded reported that it was someone other than the sheriff or police chief who was responsible for the implementation their RTCC. In almost all these cases, the individual was in a sworn position, often in middle management in the law

enforcement agency. These seven individuals were described as visionaries, always keeping up with the latest innovations in law enforcement or recognizing the value of technology.

In the four cases where the sheriff or police chief was identified as the person responsible for the implementation of the RTCC, the sheriff or police chief learned of the concept of RTCC from an outside entity. In two cases, the sheriff or police chief learned what an RTCC is through a professional association. In one instance, a sheriff learned about RTCC from a software vendor, and in another instance, a sheriff established an RTCC because a local police department already had one. In two of these cases, similar wording was used to describe the sheriff or police chief, mainly that they liked keeping up with the latest trends in law enforcement or were visionaries.

Question 14: Describe the time frame from the conception to the implementation of the RTCC.

All the agencies that responded indicated that it took 6–12 months to open their RTCC. None of the agencies reported that the opening of their RTCC was due to external factors; it was rather due to internal factors. Equipment validation and testing were frequently listed as the main reasons for delays. Another frequently listed delay was the need to establish policies and procedures on how the RTCC would function and ensuring that the RTCC complied with federal and state laws concerning data privacy. Finally, two agencies mentioned that they decided to conduct a beta test to assess the feasibility of an RTCC and that analyzing the results of the test led to delays in the implementation by several months.

Question 15: How is the RTCC primarily funded?

All the agencies informed that grant funds were used to establish their RTCCs. Commonly listed grants included the Urban Areas Security Initiative (UASI) grant and the

Edward Byrne Memorial Justice Assistance Grant. Most agencies, regardless of size, had a person at the agency designated to write and manage grants, with the largest agencies in the sample having employees that exclusively worked on grants. None of the agencies reported that they had partnered with local colleges or universities to obtain grants. Furthermore, none indicated that its RTCC was established because of the availability of grants. Finally, another popular funding source appeared to be funds obtained through asset forfeiture. Four agencies relayed that asset forfeiture funds were used to establish the agency's RTCC.

Question 16: Were community organizations' or leaders' input sought before the implementation of the RTCC?

None of the agencies that responded indicated that community organizations' or leaders' input was sought before the implementation of the RTCC. However, four of the agencies stated that advisory boards or community groups were given tours of their RTCC. One respondent mentioned as follows:

> Well, so far, we've had many community groups come in, and about three weeks ago, our chaplain said, "Hey, I got a group coming in as civilians. They're going to be very vocal." And I'm like, "Oh," he goes, "Yeah." He goes, "You know, most of the community groups are very good, but these guys you'll recognize; they're all pastors; they're very vocal in the community. They've never seen the real-time crime center, but they're very critical of the sheriff. But I want them to see the real-time crime center to show them what we're doing." I'm like, "Okay, yeah, no problem." And they came in, and they were the best group ever. They asked challenging questions, but they were like, "Well, looking at this, you need more cameras."

This trend of support emerged throughout many of the interviews and seemed to focus on using technology to produce results. Many of the respondents indicated that they are very happy to show the technology used in their RTCC to community leaders to answer any privacy questions the community may have. One respondent indicated as follows:

> I'm very cognizant of privacy issues. And that's you know, that's also the difference being the sheriff is that I can make those decisions to employ that kind of stuff, but I'm responsible to the people that I work for, you know, and if they don't like it and, you know, for years I could be in trouble. When we started out, I brought all the county commissioners in, the school superintendents, some community leaders when we started dipping our feet and the license plate readers, we did some media outreach with that, shared some success stories, you know, told people how it worked. And I've had numerous success stories, and I'm so excited about it. I've enjoyed showing it off. I invite people all the time, you know, "Hey, come by and see the crime center," and many have seen it. You know, my concern about the people being concerned that capability hasn't been there, and I have not had any pushback. They've all been very supportive of it and appreciative of the fact that we're using technology to keep them safe.

Question 17: Have other agencies contacted you regarding the implementation of an RTCC?

Eight of the 11 agencies with an RTCC that responded said they had been contacted by other agencies regarding the implementation of an RTCC. Of these eight, one agency reported that several international agencies had contacted them regarding their RTCC.

Summary

In summary, this study consisted of two distinct parts (Study 1 and Study 2). Study 1 suggested that contingency theory and resource dependency theory may be able to explain an agency's decision to establish an RTCC. Additionally, Study 1 indicated that RTCCs might be responsible for a shift in the policing model from a strictly reactive form to an enhanced reactive one that uses technology to make agencies more efficient. Study 1 also highlighted the importance technology plays in RTCCs and that this technology may be used to ease officer shortages that many agencies are experiencing.

Study 2 revealed that external factors may not be a major consideration for an agency when deciding to establish an RTCC. Furthermore, internal factors often discussed in the criminal justice literature, while present in some of the respondent's answers, may not fully explain why Florida law enforcement agencies are choosing to establish an RTCC. While Study 1 and Study 2 were able to partially explain why agencies are establishing RTCCs, further discussion is needed to delve deeper into the reasons why Florida law enforcement agencies are establishing RTCCs.

CHAPTER 4: DISCUSSION

The present study was conducted to examine the primary factors Florida law enforcement agencies used when deciding whether, or not, to establish an RTCC. However, before initiating any sort of discussion regarding the results of the study, it is important to define exactly what an RTCC is. Despite RTCCs having been in existence for over ten years, the author could not find an operational definition for an RTCC in the academic literature.

While the capabilities of RTCCs can vary widely based on agencies' budgets, personnel, and capabilities, several key features of RTCCs emerged from the data. Across the agencies, the features included the ability to data mine a wide variety of databases and then provide this data to law enforcement officers actively investigating or responding to a crime. The provided data is meant to increase the law enforcement officers' overall effectiveness while ensuring increased officer safety. Therefore, an RTCC can be operationally defined as "a unit within a police department that vets calls for service relating to crimes in progress or crimes currently being investigated and uses a centralized data hub in order to provide an effective response, increased situational awareness and enhanced officer safety." How agencies choose to define each component of this definition is up to them, and the results can vary greatly based on a multitude of contingencies.

Now that an RTCC has been defined, the relationship between the decision to establish an RTCC and the various organizational theories can be explored. Study 1 is significant because it looks at RTCCs from a perspective outside the law enforcement agency. From this perspective, external factors associated with contingency and resource dependency theory seem to be what many articles discuss. Study 2 is important because it looks at RTCCs from a perspective within

the law enforcement agency and focuses on Florida law enforcement agencies. The interviews also allow the primary researcher to focus on key concepts and themes linked to the various adaptive organizational theories. Major themes that will be discussed in Study 2 include using an RTCC to supplement for a lack of patrol officers and law enforcement's reliance on technology. These themes will be examined through the lens of an "institutional entrepreneur."

One of the findings from this study is that some law enforcement agencies have fallen out of fit due to their inability to attract and retain officers. Multiple articles reported law enforcement agencies establishing RTCCs to ease the burden of lacking patrol officers (Callaghan, 2019; Hefferman, 2021). This concept was investigated in Study 2 and was found not to be the reason why Florida law enforcement agencies are establishing RTCCs. There was a significant amount of discussion during the interviews that patrol is the most important part of the law enforcement agency and resources would be taken or diverted from specialty units to ensure patrol has adequate resources to be successful.

Another concept that emerged was the heavy reliance on technology in the operation of RTCCs. Technology was a central concept in both studies. In Study 1, vast amounts of technologies were discussed as being part of an RTCC. There were discussions of data warehouses, specialized software, facial recognition programs, and camera networks that often amounted to the hundreds or thousands. The size and scope left the impression that technology was a constraining variable and prevented almost all law enforcement agencies, except the largest, from establishing an RTCC. The focus on specialized technology placed heavy emphasis on resource dependency theory to obtain the necessary technology needed to operate an RTCC.

Study 2 shed a completely different light on the technology aspect. Technology was relevant in Study 2 but was discussed in a different way. While some of the larger agencies

reported having data warehouses and sophisticated software programs, many of the agencies discussed technology to make officers in the field more efficient. Efficiency and officer safety were commonly listed as benefits of RTCCs and how the RTCC changed the law enforcement agency. Both aspects were accomplished through technology.

For some agencies, technology did not have to be complex or sophisticated. Many agencies operated their RTCC with standard law enforcement equipment such as radios and monitor calls and standard law enforcement databases available to all law enforcement agencies. At least one medium-sized agency indicated that if it were not for the significant price decrease associated with cameras and LPRs, the widespread use of such technology would not be financially feasible. This suggests that transaction cost theory has played a role in the establishment of an RTCC for at least one agency. This agency also stated that much of the software it used was developed in-house by an employee who was very familiar with computer coding. This agency was able to leverage different non-traditional options for increasing the technology available in its RTCC. Many respondents expressed a strong desire for more cameras or license plate readers because they can beneficial but indicated that money is always a limiting factor to how quickly technology can be purchased and implemented.

While technology is important and beneficial in the operation of RTCCs, several RTCCs operated with limited technology or found ways to provide services without specialized software. This indicates that at least one of the primary reasons why Florida law enforcement agencies are creating RTCCs is to assist law enforcement officers to respond to and investigate crimes in real time. Technology enhances how RTCCs assist these officers, but some level of response and assistance can be achieved regardless of the amount of technology available in the RTCCs.

The final finding from Study 1 was that a new model of policing may need to be created instead of relying on the reactive and proactive model. During Study 2, all the respondents were asked if their RTCCs were reactive or proactive in nature. Universally, all respondents mentioned that their RTCCs were both reactive and proactive depending on the situation. This answer was not surprising given the way law enforcement officers think. For the most part, anything that an officer responds to after having been told about it by someone (dispatch, another officer, or a citizen), they will consider it reactive in nature. Anything an officer does on their own, without being told to do so, will be considered proactive in nature. Common examples include speed enforcement, running vehicle license plates looking for stolen vehicles, or providing extra patrols areas known to have high a incidence of crime. However, these actions do not meet the definition of proactive policing established by the National Academy of Sciences because the officers are reacting to or uncovering crimes that are ongoing or crimes that have already occurred (National Academy of Sciences, 2018). One respondent even said that proactive policing is not practical because it is impossible to prevent crime. This respondent indicated that the RTCC merely does what law enforcement officers have always been doing, just more efficiently and at higher volumes. This sentiment reveals that law enforcement agencies are not establishing RTCCs to switch from a reactive to proactive model, because in their minds, they have always been doing both forms of policing and the RTCC merely modernized this process and made it much more efficient. The preceding examined concepts, developed from Study 1, can explain the internal and external factors Florida law enforcement agencies were using to establish RTCCs.

Study 2 specifically dealt with the following research questions:

1) Is responding to and investigating crime in “real time” the primary reason why Florida law enforcement agencies are creating RTCCs?
2) Did external factors (e.g., elected officials, clergy, judges, community leaders, etc.) play a role in Florida law enforcement agencies’ decision to establish an RTCC?
3) Did internal factors play a role in Florida law enforcement agencies’ decision to establish an RTCC?
4) Do RTCCs stand alone, or if they interact with organizations, with whom do they interact and how frequently?

As a result of specifically dealing with the research questions, the data collected from this study specifically address each question. The results from Study 2 strongly suggest that the primary function of an RTCC is to assist law enforcement officers who are currently in the field dealing with an active crime or following up on an active crime. This would suggest that contingency theory would be a primary reason for establishing RTCCs, yet four agencies in this study were already providing some sort of assistance to officers in the field. To fight crime, at least two agencies have been using camera systems that were purchased with Homeland Security grants awarded after 9/11. Another agency had analysts monitoring radio traffic and providing analytical assistance to officers in the field way before it established an RTCC. Finally, the agency that gave an interview but did not have an RTCC acknowledged that it is already doing many of the things associated with an RTCC. These include providing analytical assistance to officers in the field, but command staff have decided that if the agency is going to establish an RTCC, it will do so with a full complement of technology. At this point, cost associated with that technology is the biggest factor for the agency establishing an RTCC.

With cost being a significant factor, do agencies with RTCCs form IORs with other organizations? This study revealed that agencies with an RTCC can form IORs with other governmental agencies. One sheriff's office has analysts from local police departments in the agency's RTCC. The respondent indicated that having analysts from various police departments facilitates the sharing of information and ensures that responding officers get access to as much data as possible. In addition to police data, the sheriff's office gets staffing support. In turn, the local police departments get access to the RTCC facility and all the technology associated with it. This arrangement would suggest that reciprocity and efficiency are the primary drivers for this sort of IOR. What is noteworthy about this relationship is that it was not a factor in the sheriff's office deciding to establish an RTCC. The IOR was formed after the RTCC was established and was an added benefit to the RTCC.

The second instance of partnerships between governmental agencies was a collaborated effort between three different agencies to form an RTCC. In this RTCC, three different agencies collaborated to form an RTCC. The sheriff's office already had an RTCC but decided to partner with the other two agencies. This decision was made for a variety of reasons, but chief among them was access for all three agencies to a bigger space, access to each agency's resources, and pooled funds that can be used to purchase upgraded technology. This partnership would also suggest that reciprocity and efficiency was a primary factor in the formation of this IOR.

Another IOR that was present in both studies was public–private partnerships between the law enforcement agency and private business that allowed access to private business camera feeds. This public–private partnership was found to be in various stages across various RTCCs that responded. Some RTCCs had robust public–private partnerships in place with MOUs that gave RTCCs access to a wealth of information, while others had a one or two businesses that had

cameras that were being monitored by the RTCC. Other RTCCs were just beginning to explore the idea of developing a public–private partnership. Regardless of how robust the public–private partnership was, this study showed that these partnerships are created after the RTCC has been established and that entering such IORs was not a reason for Florida law enforcement agencies to establish RTCCs. This finding suggests that some agencies may collaborate with other agencies to be a part of an RTCC; the vast majority of RTCCs do not consider the tenets of resource dependency theory when deciding to establish an RTCC.

The external factors and the role they played in the establishment of an RTCC will be discussed next. All the agencies that had an RTCC acknowledged that no external factors served a role in their decision to establish an RTCC, and for the most, part elected officials, clergy, and community leaders were not consulted prior to the agencies' establishment of an RTCC. This is not to say that external factors did not materialize after the RTCC was established. Multiple agencies reported giving elected officials, clergy, and community leaders tours of the RTCC. The purpose of the tours was to bring legitimacy to the RTCC and show these individuals that law enforcement organizations are being transparent regarding the operational capabilities of the RTCC. Several respondents acknowledged that certain law enforcement databases were purposefully excluded from RTCC operations to maintain the legitimacy of the RTCC. This provides strong evidence that organizational leaders are cognizant of privacy concerns and are taking active steps to ensure that the RTCC does not lose legitimacy. This finding would suggest that the concept of institutional sovereignty did not play a role in an agency's decision to establish an RTCC.

Another external factor that did not lead to agencies' establishing of RTCCs were criminal events. Criminal events are important external factors when discussing why law

enforcement agencies change. Major criminal events, such as 9/11, can lead to the development of completely new organizational concepts. One such entity created because of a criminal event were fusion centers (Taylor & Russell, 2012). In fact, the very first RTCC was created after the technology purchased because of 9/11 was repurposed to address an all-crimes approach. One would expect that at least one agency would cite major criminal events such as the school shooting at Marjorie Stoneman Douglas or the Pulse night club attack as possible reasons why agencies created their RTCC, yet no agency did. This would suggest that the external factors associated with high-profile criminal events do not factor in an agency's decision to establish an RTCC. These findings indicate that none of the external factors linked to institutional or contingency theories played a role in law enforcement agencies' decision to establish an RTCC.

Now, let us discuss the internal factors associated with an agency's decision to establish an RTCC. The first set of internal factors that are examined are associated with contingency theory. Popular internal factors of contingency theory include the size of the organization (Child, 1975) and the organization's strategy (Chandler, 1962). The results of this study found little support for organizational size or organizational strategy plays in an agency's decision to establish an RTCC. This study included many large and medium departments that established an RTCC. There were also large and medium agencies that have not established an RTCC.

Organizational strategy also did not appear to serve a significant role in an agency's decision to establish an RTCC. While many of the agencies in this study that have established an RTCC indicated that they were an ILP agency, there were also ILP agencies that have not established an RTCC. Furthermore, there were agencies that did not identify as an ILP agency that have established an RTCC and those that follow a hybrid ILP model or some other model totally different from ILP that have established an RTCC.

The next set of internal factors to be discussed are associated with institutional theory. Popular internal factors linked to institutional theory include isomorphism. DiMaggio and Powell (1991) describe three forms of isomorphism: coercive, mimetic, and normative. Coercive isomorphism was not found to be a factor in any of the agencies that were included in this study.

There was, however, an instance of normative isomorphism. A respondent reported that a sheriff created an RTCC because a police department within the county had created an RTCC. The sheriff believed it was not appropriate that the sheriff's office did not have an RTCC when a local police department had one. The respondent indicated that even though the RTCC existed, it did not function and was not successful because it was not supported by command staff and given the necessary resources it needed. No other instances of isomorphism were identified in this study.

The next internal factor associated with institutional theory is legitimacy. There were no instances where respondents suggested that their RTCC was created to increase the agency's legitimacy. As previously discussed, legitimacy was identified as being important after the RTCC was established. The internal aspect of legitimacy focused on privacy and not using computer databases that had been reported in the media as potentially violating people's privacy or having strict policies as to when cameras could be utilized. All these internal aspects only became significant after the RTCC had been established and were not factors considered when deciding to establish an RTCC.

In the criminal justice literature, institutional isomorphism and legitimacy are often cited as prominent reasons why criminal justice agencies change, yet in this study, these reasons did not fully explain why law enforcement agencies were establishing RTCCs. In this study, 53% of the agencies indicated that a sworn individual in middle management was the driving force

behind their agency establishing an RTCC. Similar phrases were often used to describe these individuals and often included words such as forward thinkers, visionaries, always keeping up with the latest innovations in law enforcement or seeing value in technology. Yet, none of the criminal justice literature reported this as reason for change. This led the primary researcher to search for an organizational concept not commonly used in criminal justice literature that could explain these middle managers being the driving force behind law enforcement agencies' establishment of an RTCC. This led to the concept of institutional entrepreneurship (DiMaggio 1988; Eisenstadt, 1980).

Eisenstadt (1980) examined the roles elites or institutional entrepreneurs play in the change that occurred in the social or political systems of traditional civilizations. DiMaggio used the idea of institutional entrepreneurs to explain how individuals can change institutions despite the environment wishing to keep things the same (Holm, 1995). Institutional entrepreneurs have been defined as individuals who bring about institutional change by leveraging resources to create or cause change within the institution (DiMaggio, 1988). Many of the early studies of institutional change failed to identify the importance of actors in organizational change. The concept of institutional actors has slowly gained popularity in the explanation of institutional change, especially concerning the change stemming from internal factors (Battilana et al., 2009).

Battilana et al. (2009) identified two conditions that must be met for an individual to be identified as an institutional entrepreneur. The first is that they must initiate change that significantly alters how the organization does business, referred to as "divergent change." Secondly, the individual must actively participate in the change. In this study, both concepts are present. This study clearly demonstrated that an RTCC involves a divergent change with several respondents identifying them as game changers. This also revealed that these individuals actively

participated in the process. Most of the institutional entrepreneurs were in a middle management position at the agency. The institutional entrepreneurs had the idea to investigate the establishment of an RTCC. That idea was presented to a member of the agency's command staff who gave permission to investigate it further or adopt the concept. The institutional entrepreneur then actively participated in a variety of ways. While not a complete list, respondents discussed visiting other RTCCs, testing and evaluating technology, identifying space, making decisions about staffing, creating policies and procedures, and planning for prolonged rollouts of the various stages of the RTCC implementation.

Another important factor represents the conditions that must be present for institutional entrepreneurs to make changes. The two key factors for ideal conditions in which institutional entrepreneurs can be successful are appropriate conditions and being in the right position within the agency (Battilana et al., 2009). In this study, both conditions were present. The concept of an RTCC is well-diffused throughout the law enforcement community, with many agencies in Florida and in other states having established them. Additionally, technology that increases efficiency and promotes officer safety is available to most departments. As previously discussed, the second condition is met by means of the middle manager being in the right position to bring about the change. While over 80% of the respondents indicated that the availability of technology was a primary factor for the establishment of their agency's RTCC, on the surface, technology appears to be the primary factor for the establishment of RTCCs. However, in almost all of these cases, a careful reading of the interview transcripts reveals the technology was acquired after the decision to explore or establish an RTCC. This suggests that in most of the agencies in this study, the institutional entrepreneur started the agency on the path to establishing an RTCC.

Limitations

This descriptive study was limited to those agencies that agreed to an interview with the primary researcher or completed the instrument and returned it to the primary researcher. As such, the small sample provides only a small snapshot into certain agencies' thinking to why they established an RTCC. Another limitation is the law enforcement agencies themselves. Many law enforcement officers have developed a unique subculture that emphasizes a different set of values that those outside the law enforcement community cannot relate to (Fuller, 2006). As a result of this subculture, many police officers are reluctant to speak with outsiders and even when they do, they tend to be careful when answering questions. The primary researcher was able to overcome many of the issues associated with the police subculture because the primary researcher is an active police officer. This relationship greatly assisted the primary researcher while conducting the interviews. Multiple respondents indicated that the interview seemed like a casual conversation between cops as opposed to an academic interview for a graduate-level degree. This sense of camaraderie however did have drawbacks. On multiple occasions, respondents indicated that since the primary researcher is a law enforcement officer, he understood what the respondent was attempting to say and could translate it into academic terms. These statements would often cause prolonged exchanges to ensure the primary researcher understood what the respondent was wishing to convey and that he will not make assumptions based on a shared profession. Another limitation of this study was the lack of interviews with agency heads.

In organizational research, the ideal candidate to interview regarding an agency's decision to establish an RTCC is the agency leader. In the case of law enforcement, that individual is the sheriff or police chief. Unfortunately, in this study, only one sheriff and one

police chief was interviewed. The letter sent to the sheriffs or police chiefs stated that the primary researcher was requesting an interview with the agency head or an individual with knowledge of the reasons as to why the agency has or has not decided to establish an RTCC. Many of the respondents told the primary researcher they had been given permission to participate in the study by the sheriff or police chief. Finally, a few respondents indicated questions they did not know the answer to and asked the sheriff or police chief for an appropriate answer. For these reasons, it was not necessary to exclude any respondents from the analysis. No other limitations were encountered during the study, and all the responding agencies cooperated in the study and placed no additional stipulations or conditions on the primary researcher.

A criticism of this study could be that it is descriptive in nature. The descriptive nature of the study may have allowed respondents to provide socially acceptable answers or answers the respondent believed the primary investigator wished to hear. Additionally, the respondents may have provided answers to questions that were not accurate or truthful as to the reasons why their agency established an RTCC. Given the research design of this study, the primary research had no option than to accept what was reported in the newspaper articles or said in the interviews. Another criticism of this study could be that a quantitative study could have been employed instead of a qualitative study. A quantitative study was considered but given the lack of research on criminal justice organizations' decision for or against establishing an RTCC, a qualitative study was ultimately selected. This decision was largely based on the ability to interact with the respondents to really capture thoughts and explore themes and concepts that organically materialize throughout the course of the interviews.

CHAPTER 5: SUMMARY AND CONCLUSION

The RTCC is a relatively new concept in the annals of law enforcement history. Spurred by tragedy, RTCCs were created by one of the largest law enforcement organizations in the United States. Since these beginnings, the concept of RTCCs has diffused across law enforcement agencies of various sizes all over the United States. This study focused on explaining why law enforcement agencies are creating RTCCs. Interviews with law enforcement personnel were conducted, and with a set of answers to questions, organizational theories dealing with adaptive changes were applied to these answers to see if any of the theories could explain why agencies are establishing RTCCs.

To bring a level of order to the data, possible reasons from each theory were further divided into internal and external factors. This study showed that internal and external factors such as institutional pressures, agency legitimacy, adapting to task environments, isomorphism, or contingencies like agency size or budget could not fully explain an agency's decisions to establish an RTCC. Furthermore, popular reasons for establishing IORs such as reciprocity and efficiency (Oliver, 1990) were discussed in detail during the interviews but were often cited as a benefit of the RTCC and not as a reason for its establishment.

The unique finding of this study was "agency champions" represent the biggest cause for the establishment of RTCCs. The idea that "agency champions" are a reason for change did not appear in the criminal justice literature reviewed for this study. As a result, when it became apparent that middle managers were mostly responsible for the establishment of RTCCs in many of the organizations included in this study, a search for an explanation was begun. What was found was a term in criminal justice literature, namely institutional entrepreneurship, which had not been

previously researched in the criminal justice peer-reviewed literature. The idea suggests that an individual can see an emerging trend, see the value in that trend, take ownership of the idea, and work toward implementing the idea at the law enforcement agency. This describes exactly what was noted in many of the interviews. Individuals who saw the potential that having technology could increase officer safety, make the law enforcement agency more efficient, and solve crimes that in the past may never have been solved subsequently decided to work with their chain of command to establish their agency's RTCC.

Future Research

Organizational research on RTCCs is not very well established. This study was meant to provide a foundational explanation as to why Florida law enforcement agencies are establishing RTCCs. Based on the findings of this study, it is apparent that more research into RTCCs is necessary. In this study, only two agencies that currently do not have RTCCs provided data. As of this writing, there were 19 sheriff's offices that had RTCCs and 48 Florida sheriff's offices that did not. These agencies can provide a counterpoint to the agencies that participated in this study. There are also myriad possibilities of the effectiveness of RTCCs. RTCCs are expensive, and if they are not keeping officers safe or providing added benefits to law enforcement agencies, then law enforcement agencies' heads should be made aware of that. Finally, agencies would benefit if they understood what effects or deterrents RTCCs have on crime, if any. All these questions have been researched with other aspects of law enforcement programs, and there is no reason to believe that RTCCs would not benefit from the same level of research.

APPENDIX A: LIST OF TABLES

Table 1 Summary of Adaptive Organizational Theories

Theory	Main Reason for Change		
		Internal Factors	External Factors
Contingency Theory	Agencies must adapt to changes in the environment in order to ensure fit. Fit ensures agency effectiveness	Organizational Strategy Organizational Size Organizational Jurisdiction	Changes in the Environment Changes in Tasks Changes in Technology
Resource Dependency Theory	In order to be seen as legitimate agencies must partner with other agencies to acquire needed resources. Those with control over the resource are seen as more legitimate and has having more power over the agency seeking the resource	Resource Dependency Theory is all about external factors	Changes in environment Changes in Power Structure Uncertainty of Resources
Institutional Theory	To gain legitimacy organizations acquire must acquire resources; however, unlike resource dependency theory there are internal factors that drive the need to acquire resources	To become more like other agencies (isomorphism) Change brought about by agency personnel	Change in order to appear more legitimate to stakeholders

Table 2 Study 2 Description of Survey Recipients and Respondents

Type of Law Enforcement Agency	Survey Recipients *N*	Survey Respondents *n (%)*
Sheriff's Offices with an RTCC	18	7 (38%)
Sheriff's Offices without an RTCC	20	2 (10%)
Police Departments with an RTCC	11	4 (36%)
University Police Departments With an RTCC	1	0 (0%)
University Police Departments Without an RTCC	3	0 (0%)
Total	53	13 (24%)

APPENDIX B: SURVEY INSTRUMENT

1) Would you consider your agency to be an intelligence led policing Agency?

   For this interview intelligence led policing will be defined as: "[t]he collection and analysis of information related to crime and conditions that contribute to crime, resulting in an actionable intelligence product intended to aid law enforcement in developing tactical responses to threats and/or strategic planning related to emerging or changing threats (Carter, 2009, p.13)"[1]

   Probing Question: What do you particularly like about being an intelligence led policing agency?

2) Did your law enforcement agency have an intelligence component prior to the establishment of a RTCC?

3) If not, was the intelligence unit created when the RTCC was established?

   Probing Question: Why do you feel it was necessary to create an intelligence unit in addition to the RTCC?

4) If the agency had an intelligence unit, was it rebranded as an RTCC?

   Probing Question: Why was rebranding your intelligence unit as an RTCC the right decision for your agency?

5) If so, does the intelligence unit still exist?

---

[1] U.S. Department of Justice, Office of Community Oriented Policing Services, *Law Enforcement Intelligence: A Guide for State, Local, and Tribal Law Enforcement Agencies*, David L. Carter, 2nd ed. (May, 2009): 80, accessed February 7, 2022, *https://it.ojp.gov/documents/d/e050919201-IntelGuide_web.pdf;* Ibid; and Thomas E. Baker, *Intelligence-Led Policing* (Flushing, NY: Looseleaf Law Publications, 2009): 43.

a. Was the intelligence Unit added under the umbrella of the RTCC?

b. If the intelligence unit was rebranded as an RTCC what steps were done to create an RTCC?

c. If the intelligence unit still exists how has the Intelligence Unit mission changed with the addition of a RTCC?

d. Explain what has stayed the same.

6) Was the primary reason the RTCC was established a criminal event?

   a. If so, what was the nature of the event?

   b. What was significant about this event that your agency felt it was necessary to create a RTCC?

7) If the primary reason for the creation of the RTCC was something other than a criminal event, could you explain what was the reason? (Contingency Theory)

   Probing Question: Could you briefly explain why your agency felt it was necessary to establish a RTCC?

8) What are the primary functions of the RTCC?

   a. What is the composition of the staff that work in the RTCC?

      i. If sworn law enforcement work in the RTCC what is their primary job?

      ii. Do analyst complete requests for other sections of the agency?

      iii. Do analyst complete requests for other agencies?

9) How has a RTCC changed the way your agency operates?

   a. Would you consider this change a good change or bad change?

Probing Question: Could you give me an example of how having an RTCC has changed the way your agency operates?

10) What benefits have you seen as a result of the establishment of a RTCC?

Probing Question: Could you elaborate on these benefits and how they have improved your agency?

11) What are some negative aspects of establishing a RTCC?

Probing Question: Do you feel that these negative aspects have had a major impact on how your agency operates?

12) Does your agency keep records of RTCC success stories or statistics on the number of criminal events in which RTCC resources were utilized?

    a. Have you seen a significant use in the resources or capabilities offered by the RTCC?

Probing Question: Do you feel that the significant use of these resources or capabilities has helped or hurt your agency?

13) Who in your agency was the primary person/role responsible for the implementation of the RTCC? (Institutional Theory, Sovereigns)

    a. How did they become aware of the concept of a RTCC?

        i. If a conference or trade show what was the name?

        ii. If it was another agency, what was the agency?

        iii. Was it another state?

14) Describe the timeframe from conception to implementation of the RTCC? (Resource Dependency Theory)

    a. Did the Sheriff or Police Chief seek outside approval for the creation of the RTCC

        i. If so, from who? (Board of County Commission or Mayor's Office)?

    b. What was the time frame from conception to implementation?

        i. If there was a delay what was the cause of the delay?

15) How is the RTCC primarily funded? (Resource Dependency Theory)

    a. Were grant funds used to pay for the RTCC?

    b. Was the RTCC established because of the availability of grant funds?

    c. If so, who wrote/what role wrote the grant?

    d. (If external) Had you used this organization to write previous grants?

16) Were community organizations or leaders' input sought before the implementation of the RTCC? (Institutional Theory)

    a. If community organizations or leaders' input was sought before the implementation of a RTCC why was their input sought?

    b. To the best of your recollection did the entity express support or reservations about the creation of the RTCC?

        i. If the organization expressed reservations what were the reservations?

Probing Question: Could you elaborate on why you thought it was necessary to seek input form the community regarding the establishment of the RTCC?

17) Have other agencies contacted you regarding the implementation of a RTCC? If yes: (Institutional Theory, Isomorphism)

a. Are they located in Florida?

b. If located outside of Florida, what state are they located?

18) Is there anything else that I have not asked you that you feel is important regarding your agency's decision to establish a RTCC?

APPENDIX C: IRB APPROVAL LETTER

UNIVERSITY OF CENTRAL FLORIDA

Institutional Review Board
FWA00000351
IRB00001138, IRB00012110
Office of Research
12201 Research Parkway
Orlando, FL 32826-3246

NOT HUMAN RESEARCH DETERMINATION

March 25, 2022

Dear Frank Linton:

On 3/25/2022, the IRB reviewed the following protocol:

Type of Review:	Initial Study, program evaluation
Title of Study:	Real-Time Crime Centers: A Mixed Model Study of Why Law Enforcement Agencies are Establishing Them
Investigator:	Frank Linton
IRB ID:	STUDY00004074
Funding:	None
Grant ID:	None
Documents Reviewed:	• Interview Questions, Category: Interview / Focus Questions; • UCF Form 250, Category: IRB Protocol;

The IRB determined that the proposed activity is not research involving human subjects as defined by DHHS and FDA regulations.

IRB review and approval by this organization is not required. This determination applies only to the activities described in the IRB submission and does not apply should any changes be made. If changes are made and there are questions about whether these activities are research involving human in which the organization is engaged, please submit a new request to the IRB for a determination. You can create a modification by clicking **Create Modification / CR** within the study.

If you have any questions, please contact the UCF IRB at 407-823-2901 or irb@ucf.edu. Please include your project title and IRB number in all correspondence with this office.

Sincerely,

UNIVERSITY OF CENTRAL FLORIDA

Institutional Review Board
FWA00000351
IRB00001138, IRB00012110
Office of Research
12201 Research Parkway
Orlando, FL 32826-3246

Harry Wingfield
UCF IRB

REFERENCES

Adams, W. C. (2015). Conducting semi-structured interviews. In J. S. Wholey, H. P. Harty, & K. E. Newcomer (Eds.), *Handbook of Practical Program Evaluation* (pp. 492–505). Jossey-Bass. https://doi.org/10.1002/9781119171386.ch19

Aldrich, H. (1999). *Organizations evolving*. Sage Publications.

Audit Commission. (1993). *Helping with enquiries: Tackling crime effectively*. (Management handbook). HMSO.

Barisal, L., & While, A. (1994). Collecting data using a semi-structured interview: A discussion paper. *Journal of Advanced Nursing*, *19*(2), 328–335. https://doi.org/10.1111/j.1365-2648.1994.tb01088.x

Battilana, J., Leca, B., & Boxenbaum, E. (2009). How actors change institutions: Towards a theory of institutional entrepreneurship. *Academy of Management Annals*, *3*(1), 65–107. https://doi.org/10.5465/19416520903053598

Benson, J. K. (1975). The interorganizational network as a political economy. *Administrative Science Quarterly*, *20*(2), 229. https://doi.org/10.2307/2391696

Braga, A. A., & Weisburd, D. (2006). Critic Problem–oriented policing: The disconnect between principles and practice. *Police Innovation*, 133–152. https://doi.org/10.1017/cbo9780511489334.007

Brewer, T. W., Jefferis, E., Butcher, F., & Wiles, T. D. (2007). A case study of the Northern Ohio Violent Fugitive Task Force. *Criminal Justice Policy Review*, *18*(2), 200–220. https://doi.org/10.1177/0887403406295310

Bullock, K. (2013). Community, intelligence-led policing and crime control. *Policing and Society*, *23*(2), 125–144. https://doi.org/10.1080/10439463.2012.671822

Bundy, M. G. (1970). Press Conference, Presented in New York City, NY.

Burns, T., & Stalker, G. M. (1961). *The management of innovation*. Tavistock Publications.

Callaghan, A. (2019, May 6). *Safecam to assist in New Orleans crime reduction*. The Maroon. Retrieved July 13, 2022, from https://loyolamaroon.com/10022906/worldview/city/safecam-to-assist-in-new-orleans-crime-reduction/

Carter, D. L., & Carter, J. G. (2008). Intelligence-led policing. *Criminal Justice Policy Review*, *20*(3), 310–325. https://doi.org/10.1177/0887403408327381

Carter, D. L., & Carter, J. G. (2009). The intelligence fusion process for state, local, and tribal law enforcement. *Criminal Justice and Behavior*, *36*(12), 1323–1339. https://doi.org/10.1177/0093854809345674

Carter, J. G. (2016). Institutional pressures and isomorphism. *Police Quarterly*, *19*(4), 435–460. https://doi.org/10.1177/1098611116639536

Carter, J., & Chermak, S. (2011). Evidence-based intelligence practices: Examining the role of fusion centers as a critical source of information. In C. Lum & L. Kennedy (Eds.), *Evidence-Based Counterterrorism Policy* (pp. 65–88). Springer. https://doi.org/10.1007/978-1-4614-0953-3_4

Chandler, A. D. (1962). *Strategy and structure: Chapters in the history of the American industrial enterprise*. Beard Book.

Child, J. (1975). *On organizations in their sectors – John Child, 1988*. https://journals.sagepub.com/doi/10.1177/017084068800900105

City of New York. (2010, April 27). *Best practice: Real time crime center: centralized crime data system*. New York City Global Partners. http://www.nyc.gov/html/unccp/gprb/downloads/pdf/NYC_Policy%20and%20Procedure%20Recommendation%20Program.pdf?epi-content=GENERIC

The COMPSTAT Office. (n.d.). *Office of Management Analysis and Planning New York: NYPD*.

Corbin, J., & Strauss, A. L. (2015). *Basics of qualitative research: Techniques and procedures for developing grounded theory*. Sage.

Cordner, G. W., & Sheehan, R. (1999). *Police administration*. Routledge.

Crank, J. P., & Langworthy, R. (1992). An institutional perspective of policing. *The Journal of Criminal Law & Criminology*, *85*(2).

Cridland, E. K., Jones, S. C., Caputi, P., & Magee, C. A. (2014). Qualitative research with families living with autism spectrum disorder: Recommendations for conducting semistructured interviews. *Journal of Intellectual and Developmental Disability*, *40*(1), 78–91. https://doi.org/10.3109/13668250.2014.964191

Damanpour, F. (1991). Organizational innovation: A meta-analysis of effects of determinants and moderators. *Academy of Management Journal*, *34*(3), 555–590. https://doi.org/10.5465/256406

Department of Homeland Security. (2021, September 30). *National Network of Fusion Centers fact sheet*. Department of Homeland Security. https://www.dhs.gov/national-network-fusion-centers-fact-sheet

Devine, T. (2014). *An examination of the effectiveness of state and local fusion centers toward federal counterterrorism efforts* [Thesis]. INSS Capstone, El Paso.

DiCicco-Bloom, B., & Crabtree, B. F. (2006). The qualitative research interview. *Medical Education*, *40*(4), 314–321.

DiMaggio, P. J. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, *48*, 147–60. *Organizational Sociology*, 517–530. https://doi.org/10.4324/9781315247533-34

DiMaggio, Paul J. (1988). Interest and agency in institutional theory. In L. Zucker (Ed.), *Institutional patterns and organizations* (pp. 3–22). Ballinger.

Donaldson, L. (2001). *The contingency theory of organizations*. Sage Publications.

Eisenstadt, S. N. (1980). Cultural orientations, institutional entrepreneurs, and social change: Comparative analysis of traditional civilizations. *American Journal of Sociology*, *85*(4), 840–869. https://doi.org/10.1086/227091

Elo, S., Kääriäinen, M., Kanste, O., Pölkki, T., Utriainen, K., & Kyngäs, H. (2014). Qualitative content analysis. *SAGE Open*, *4*(1), 215824401452263. https://doi.org/10.1177/2158244014522633

Freeman, J., & Hannan, M. T. (1989). Setting the record straight on organizational ecology: Rebuttal to young. *American Journal of Sociology*, *95*(2), 425–439. https://doi.org/10.1086/229275

Fuller, J. R. (2006). *Criminal justice mainstreams and crosscurrents*. Prentice Hall.

German, M., & Stanley, J. (n.d.). *Fusion center update – ACLU*. Fusion Center Update. https://www.aclu.org/sites/default/files/pdfs/privacy/fusion_update_20080729.pdf

Giblin, M. J., & Burrus, G. W. (2009). Developing a measurement model of institutional processes in policing. *Policing: An International Journal of Police Strategies & Management*, *32*(2), 351–376. https://doi.org/10.1108/13639510910958226

Greenwood, R., & Hinings, C. R. (1988). Organizational design types, tracks and the dynamics of strategic change. *Organization Studies*, *9*(3), 293–316. https://doi.org/10.1177/017084068800900301

Hall, R. H., & Zucker, L. G. (1989). Institutional patterns and organizations: Culture and environment. *Contemporary Sociology*, *18*(1), 54. https://doi.org/10.2307/2071932

Hannan, M. T., & Freeman, J. (1977). The population ecology of organizations. *American Journal of Sociology*, *82*(5), 929–964.

Hayeslip, D., & Russell-Einhorn, M. L. (2002). *Evaluation of multi-jurisdictional task forces project: Phase I final report*. National Criminal Justice Reference Service (NCJRS).

Heaton, R. (2000). The prospects for intelligence-led policing: Some historical and quantitative considerations. *Policing and Society*, *9*(4), 337–355. https://doi.org/10.1080/10439463.2000.9964822

Heffernan, E. (2022, January 19). *St. Louis progressive groups campaign to 'defund' St. Louis Police in next city budget*. STLtoday.com. Retrieved July 13, 2022, from https://www.stltoday.com/news/local/metro/st-louis-progressive-groups-campaign-to-defund-st-louis-police-in-next-city-budget/article_242ab3d6-afc8-5d4f-8b0d-b416d6e5e150.html

Henry, V. E. (2003). *The COMPSTAT paradigm: Management accountability in policing, business, and the public sector*. Looseleaf Law Publications.

Hollenbeck, J. R., Moon, H., Ellis, A. P., West, B. J., Ilgen, D. R., Sheppard, L., Porter, C. O., & Wagner, J. A. (2002). Structural contingency theory and individual differences: Examination of external and internal person-team fit. *Journal of Applied Psychology*, *87*(3), 599–606. https://doi.org/10.1037/0021-9010.87.3.599

Holm, P. (1995). The dynamics of institutionalization: Transformation processes in Norwegian fisheries. *Administrative Science Quarterly*, *40*(3), 398. https://doi.org/10.2307/2393791

Hsieh, H.-F., & Shannon, S. E. (2005). Three approaches to qualitative content analysis. *Qualitative Health Research*, *15*(9), 1277–1288. https://doi.org/10.1177/1049732305276687

Jamshed, S. (2014). Qualitative research method – Interviewing and observation. *Journal of Basic and Clinical Pharmacy*, *5*(4), 87. https://doi.org/10.4103/0976-0105.141942

Kallio, H., Pietilä, A-M., Johnson, M., & Kangasniemi, M. (2016). Systematic methodological review: Developing a framework for a qualitative semi-structured interview guide. *Journal of Advanced Nursing*, *72*(12), 2954–2965. https://doi.org/10.1111/jan.13031

Katz, C. M., Maguire, E. R., & Roncek, D. W. (2002). The creation of specialized police gang units. *Policing: An International Journal of Police Strategies & Management*, *25*(3), 472–506. https://doi.org/10.1108/13639510210437005

Kelling, G. L., & Coles, C. M. (1996). *Fixing broken windows: Restoring order and reducing crime in our communities*. Touchstone.

King, W. R. (2000). Measuring police innovation: Issues and measurement. *Policing: An International Journal of Police Strategies & Management*, *23*(3), 303–317. https://doi.org/10.1108/13639510010342994

Kitto, S. C., Chesters, J., & Grbich, C. (2008). Quality in qualitative research. *Medical Journal of Australia*, *188*(4), 243–246. https://doi.org/10.5694/j.1326-5377.2008.tb01595.x

Kondracki, N. L., Wellman, N. S., & Amundson, D. R. (2002). Content analysis: Review of methods and their applications in nutrition education. *Journal of Nutrition Education and Behavior*, *34*(4), 224–230. https://doi.org/10.1016/s1499-4046(06)60097-3

Lawrence, P. R., & Lorsch, J. W. (1967). Differentiation and integration in complex organizations. *Administrative Science Quarterly*, *12*(1), 1. https://doi.org/10.2307/2391211

Magers, J. S. (2004). COMPSTAT: A new paradigm for policing or a repudiation of community policing. *Journal of Contemporary Criminal Justice*, *20*(1), 70–79. https://doi.org/10.1177/1043986203262312

Maguire, E. R. (2014). Police organizations and the iron cage of rationality. *Oxford Handbooks Online*. https://doi.org/10.1093/oxfordhb/9780199843886.013.019

Maple, J., & Mitchell, C. (1999). *The crime fighter: Putting the bad guys out of business*. Doubleday.

Massee, T., & Rollins, J. (2007, September 19). *A summary of fusion centers: Core issues and options for Congress.* https://sgp.fas.org/crs/intel/RL34177.pdf

Matusiak, M. C. (2018). Environmental predictors of municipal police agency goals. *Police Quarterly*, *22*(1), 112–136. https://doi.org/10.1177/1098611118797068

Maurer, J. G. (1971). *Readings in organization theory: Open-system approaches*. Random House.

McDonald, P. (2004). *CompStat and beyond*. Policechiefmagazine.org. http://policechiefmagazine.org/magazine/index.cfm?fuseaction=display&article_id=202&issue_id-12004

McDonald, P. P., & Greenberg, S. F. (2002). *Managing police operations: Implementing the New York crime control model COMPSTAT*. Wadsworth Pub.

McElhaney, B. (2019). *Advancing collaboration between joint terrorism task forces and fusion centers* [Dissertation]. ProQuest.

Meyer, J. W., & Rowan, B. (1977). Institutionalized organizations: Formal structure as myth and ceremony. *American Journal of Sociology*, *83*(2), 340–363. https://doi.org/10.1086/226550

Meyer, J. W., & Scott, W. R. (1992). *Organizational environments: Ritual and rationality*. Sage Publications.

Meyer, M. W. (1978). *Environments and organizations*. Jossey-Bass.

Monteverde, K., & Teece, D. J. (1982). Supplier switching costs and vertical integration in the automobile industry. *The Bell Journal of Economics*, *13*(1), 206. https://doi.org/10.2307/3003441

National Academy of Sciences. (2018). *Proactive policing: Effects on crime, communities, and civil liberties in the United States*. https://www.nationalacademies.org/our-work/proactive-policing-effects-on-crime-communities-and-civil-liberties-in-the-united-states

Nelson, R. R., & Winter, S. G. (2004). *An evolutionary theory of economic change*. The Belknap Press of Harvard University Press.

New York City Global Partners Innovation Exchange. (2006). *Best Practice: Real Time Crime Center: Centralized Crime Data System*. http://www.nyc.gov/html/unccp/gprb/downloads/pdf/NYC_Safety%20and%20Security_RTCC.pdf

Oliver, C. (1990). Determinants of interorganizational relationships: Integration and future directions. *The Academy of Management Review*, *15*(2), 241. https://doi.org/10.2307/258156

Oliver, W. M. (2006). The fourth era of policing: Homeland security. *International Review of Law, Computers & Technology*, *20*(1-2), 49–62. https://doi.org/10.1080/13600860600579696

Parsons, T. (1956). Suggestions for a sociological approach to the theory of organizations. *Administrative Science Quarterly*, *1*(1), 63. https://doi.org/10.2307/2390840

Pennings, J. M. (1987). Structural contingency theory: A multivariate test. *Organization Studies*, *8*(3), 223–240. https://doi.org/10.1177/017084068700800302

Pennings, J. M. (1992). *Structural contingency theory: A re-appraisal.* (Vol. XIV). JAI Press.

Perrow, C. (1967). A framework for the comparative analysis of organizations. *American Sociological Review*, *32*(2), 194. https://doi.org/10.2307/2091811

Pfeffer, J., & Salancik, G. R. (2003). *The external control of organizations: A resource dependence perspective*. Stanford Business Classics.

Ratcliffe, J.H. (2016). *Intelligence-led policing*. Routledge, Taylor & Francis Group.

Roy, C., Séguin, F., (2000). The institutionalization of efficiency-oriented approaches for public service improvement. *Public Productivity & Management Review*, *23*(4), 449. https://doi.org/10.2307/3380563

Safir, H. (n.d.). *The COMPSTAT process*. Office of Management Analysis and Planning New York: NYPD.

Sarta, A., Durand, R., & Vergne, J-P. (2020). Organizational adaptation. *Journal of Management*, *47*(1), 43–75. https://doi.org/10.1177/0149206320929088

Schaible, L. M., & Sheffield, J. (2012). Intelligence-led policing and change in state law enforcement agencies. *Policing: An International Journal of Police Strategies & Management*, *35*(4), 761–784. https://doi.org/10.1108/13639511211275643

Scott, W. R. (2014). *Institutions and organizations: Ideas and interests*. Sage Publications.

Scott, W. R., & Davis, G. F. (2017). *Organizations and organizing: Rational, natural, and open system perspectives*. Routledge.

Shearing, C., & Wood, J. (2007). From the guest editors: Re-shaping policing: Ideas in action. *Police Practice and Research*, *8*(2), 99–106. https://doi.org/10.1080/15614260701377570

Sheehan, R., Cordner, G. W., & Sheehan, R. (1995). *Police administration*. Anderson Publishing.

Sherman, L. W., & Weisburd, D. (1995). General deterrent effects of police patrol in crime "hot spots": A randomized, controlled trial. *Justice Quarterly*, *12*(4), 625–648. https://doi.org/10.1080/07418829500096221

Suchman, M. C. (1995). Managing legitimacy: Strategic and institutional approaches. *Academy of Management Review*, *20*(3), 571–610. https://doi.org/10.5465/amr.1995.9508080331

Sung, H-E., Capellan, J., & Barthuly, B. (2022). Trust in the police and the militarization of law enforcement in Latin America. *Policing and Society*, *32*(3), 311–340. https://doi.org/10.1080/10439463.2022.2037551

Taylor, R. W., & Russell, A. L. (2012). The failure of police "fusion" centers and the concept of a national intelligence sharing plan. *Police Practice and Research*, *13*(2), 184–200. https://doi.org/10.1080/15614263.2011.581448

United States Department of Justice. (2006, August). *Fusion center guidelines: Developing and sharing information and intelligence in a new era*. Global Justice Information Sharing initiative.

https://www.iadlest.org/Portals/0/Files/Documents/DDACTS/Docs/Fusion%20Center%20Guidelines%20-%20Executive%20Summary,%20DOJ,%20August%202005.pdf

Van de Ven, A., & Drazin, R. (1985). *The concept of fit in contingency theory*. Strategic Management Research Center.

Vito, G. F., Walsh, W. F., & Kunselman, J. (2005). COMPSTAT: The manager's perspective. *International Journal of Police Science & Management*, *7*(3).

Waddington, P.A. (1993). *Calling the police: The interpretation of, and response to, calls for assistance from the public*. Avebury.

Walsh, W. F. (2001). COMPSTAT: An analysis of an emerging police managerial paradigm. *Policing: An International Journal of Police Strategies & Management*, *24*(3), 347–362. https://doi.org/10.1108/13639510110401717

Weisburd, D., Telep, C. W., Hinkle, J. C., & Eck, J. E. (2008). The effects of problem-oriented policing on crime and disorder. *Campbell Systematic Reviews*, *4*(1), 1–87. https://doi.org/10.4073/csr.2008.14

Wiewel, W., & Hunter, A. (1985). The interorganizational network as a resource: A comparative case study on organizational genesis. *Administrative Science Quarterly*, *30*(4), 482. https://doi.org/10.2307/2392693

Williamson, O.E. (1985). *The economic institutions of capitalism*. Free Press.

Williamson, O. E. (1975). Transaction cost economics: An overview. In P. G. Klein & M. E. Sykuta (Eds.), *The Elgar companion to transaction cost economics*. Edward Elgar Publishing Ltd. https://doi.org/10.4337/9781849806909.00007

Willis, J. J., Mastrofski, S. D., & Weisburd, D. (2007). Making sense of COMPSTAT: A theory-based analysis of organizational change in three police departments. *Law & Society Review*, *41*(1), 147–188. https://doi.org/10.1111/j.1540-5893.2007.00294.x

Woodward, J., Dawson, S., & Wedderburn, D. (1965). *Industrial organization: Theory and practice*. Oxford University Press.

Zhao, J., Ren, L, & Lovrich, N. (2010). Police organizational structures during the 1990s: An application of contingency theory. *Police Quarterly*, *13*(2), 209–232. https://doi.org/10.1177/1098611110365691